

#4 REBAR FOR VERTICALS
COLUMN GROUTED SOLID

16" 8" 8"

1-#4 HORIZONTAL REBAR

NATURAL GRADE OR TOP OF IRRIGATION BERM

12' O.C.

6'0"

1'0"

PROVIDE 8"x16" DRAINAGE OPENINGS, EVERY OTHER BLOCK. NO DECORATIVE BLOCK ALLOWED.

NTS

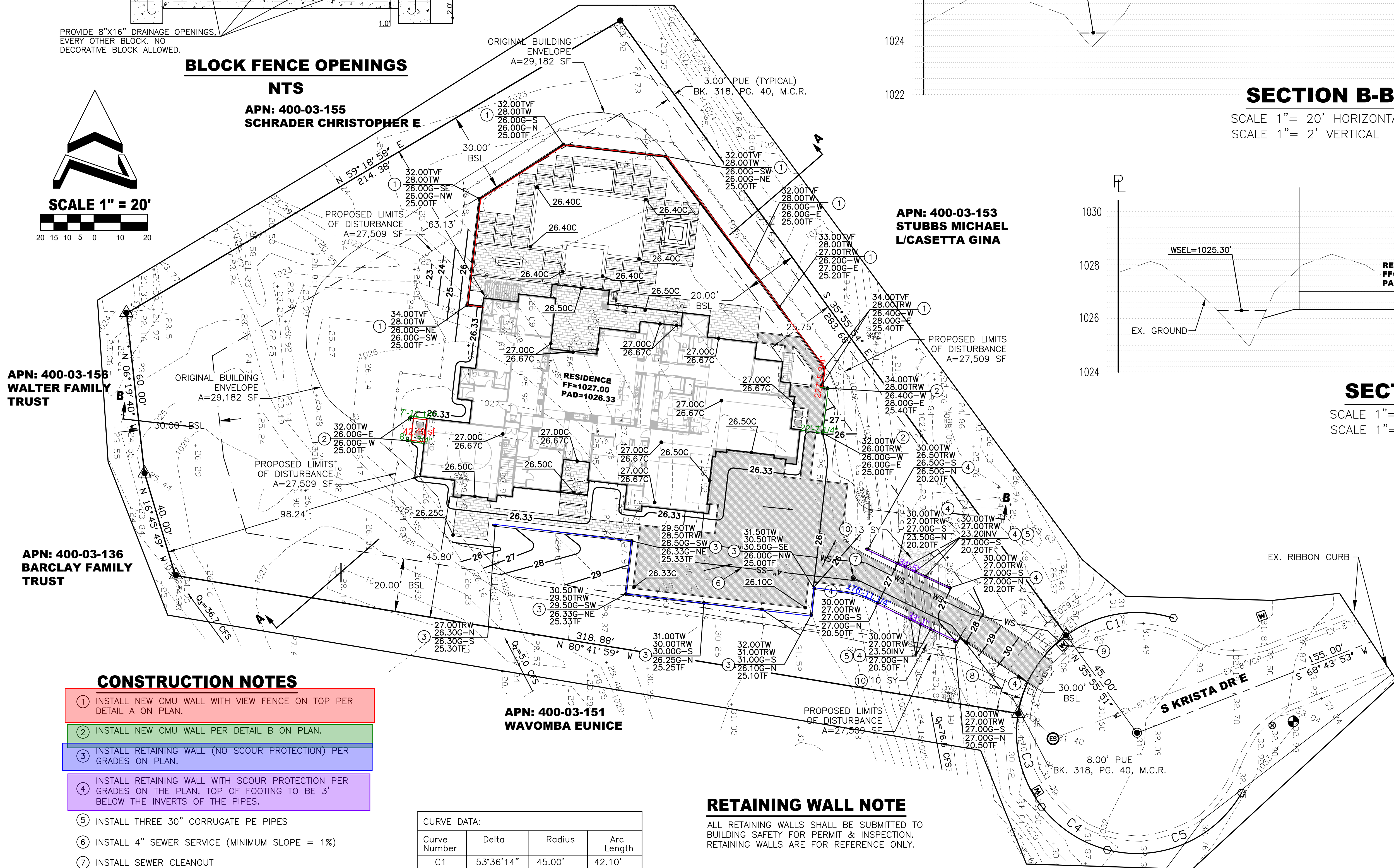


SCALE 1" = 20'



- BRASS CAP
- SET 1/2" REBAR LS#41076
- ▲ FOUND 1/2" REBAR
- NOTHING FOUND/NOTHING SET
- ⊗ FIRE HYDRANT
- ⊗ WATER VALVE
- W WATER SERVICE/WATER METER
- ES EXISTING SEWER MANHOLE
- J TELECOM JUNCT. BOX
- E ELECTRICAL TRANSFORMER

	PALO VERDE
	SAGUARO CACTUS
BSL	BUILDING SETBACK LINE
PUE	PUBLIC UTILITY EASEMENT
M.C.R.	MARICOPA COUNTY RECORDER
TRW	TOP OF RETAINING WALL
TVF	TOP OF VIEW FENCE
TW	TOP OF WALL
TF	TOP OF FOOTING
G	GROUND
C	CONCRETE



- ① INSTALL NEW CMU WALL WITH VIEW FENCE ON TOP PER DETAIL A ON PLAN.
- ② INSTALL NEW CMU WALL PER DETAIL B ON PLAN.
- ③ INSTALL RETAINING WALL (NO SCOUR PROTECTION) PER GRADES ON PLAN.
- ④ INSTALL RETAINING WALL WITH SCOUR PROTECTION PER GRADES ON PLAN. TOP OF FOOTING TO BE 3' BELOW THE INVERTS OF THE PIPES.

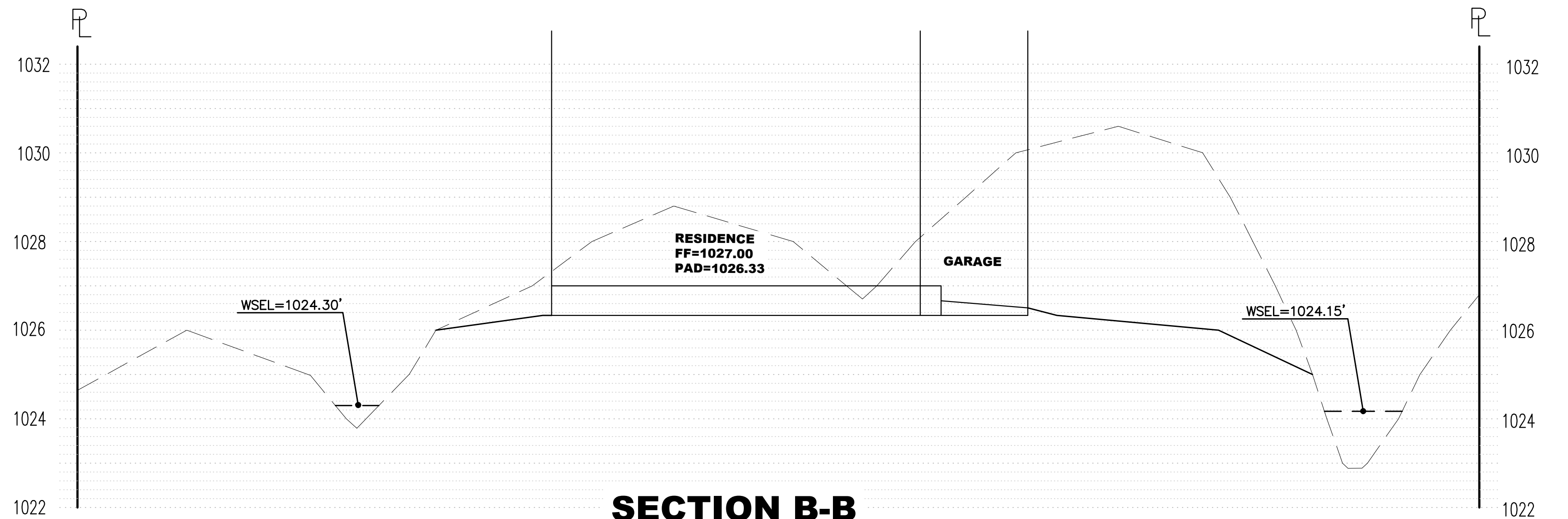
- ⑤ INSTALL THREE 30" CORRUGATE PE PIPES
- ⑥ INSTALL 4" SEWER SERVICE (MINIMUM SLOPE = 1%)
- ⑦ INSTALL SEWER CLEANOUT
- ⑧ THE SEWER STUB IS SHOWN AS EXISTING ON THE AS-BUILT SEWER MAPS. CONTRACTOR TO VERIFY.
- ⑨ REPLACE EX 3/4" WATER METER WITH 1" WATER METER AND WATER SERVICE.

⑩ INSTALL D50=8" HANDPLACED ANGULAR STONE RIP RAP OVER AASHTO M288 CLASS 2 NON WOVEN GEO-TEXTILE FILTER FABRIC OR APPROVED EQUAL AT EACH LOCATION THICKNESS=16".

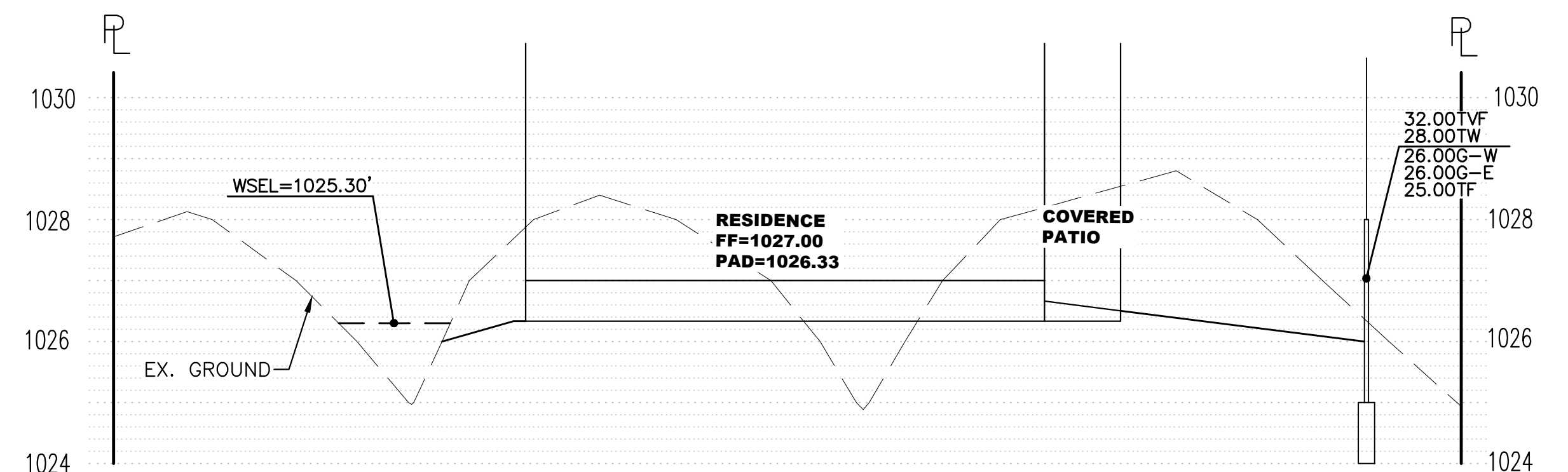
CURVE DATA:			
Curve Number	Delta	Radius	Arc Length
C1	53°36'14"	45.00'	42.10'
C2	44°46'09"	45.00'	35.16'
C3	47°00'45"	45.00'	36.92'
C4	38°08'03"	45.00'	29.95'
CX	74°22'07"	45.00'	58.41'

ALL RETAINING WALLS SHALL BE SUBMITTED TO
BUILDING SAFETY FOR PERMIT & INSPECTION.
RETAINING WALLS ARE FOR REFERENCE ONLY.

ALL PROPOSED PERIMETER WALLS, RETAINING WALLS, AND POOL LOCATIONS SHALL BE SUBMITTED TO THE CITY OF GOODYEAR ENGINEERING DEPARTMENT FOR REVIEW OF DRAINAGE ISSUES PRIOR TO CONSTRUCTION. ANY WALL CONSTRUCTED WITHOUT PRIOR APPROVAL MAY BE SUBJECT TO REMOVAL.



SCALE 1"= 20' HORIZONTAL
SCALE 1"= 2' VERTICAL



SCALE 1"= 20' HORIZONTAL
SCALE 1"= 2' VERTICAL

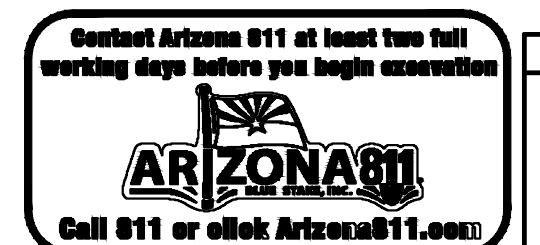
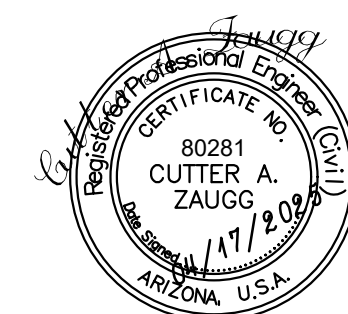
Technical drawing of a 6' high CMU wall. The wall is constructed of CMU blocks with #4 horizontal rebar. A 30% transparent screen is attached to the exterior face of the wall. The screen has vertical bars spaced at 1" and horizontal bars spaced at 4" MAX. The wall is 6' high. The base of the wall is at the natural grade or top of the irrigation berm. The wall is supported by a concrete footing. The footing is 18" wide and has a 3" thick layer of concrete above it. The footing is labeled "CONC. FOOTING".

Labels and dimensions include:

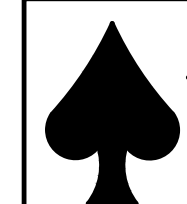
- 6' (Total wall height)
- 30% TRANSPARENT (Screen label)
- 1" (Vertical screen bar spacing)
- 4" MAX (Horizontal screen bar spacing)
- CMU WALL
- #4 HORIZONTAL REBAR
- 2.0' (Height from footing to top of wall)
- NATURAL GRADE OR TOP OF IRRIGATION BERM
- 18" (Footing width)
- 3" (Concrete layer above footing)
- CONC. FOOTING
- 1-#4 REBAR (Vertical rebar in footing)

PROVIDE 8"x16" DRAINAGE OPENINGS AT NATURAL GRADE EVERY OTHER BLOCK.

N.T.S.



DATE _____



**ALLEN
CONSULTING
ENGINEERS, P.C.**

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MESA, ARIZONA 85206
PHONE (480) 844-1666
E-MAIL: ace@allenconsultengr.com

APN: 400-03-152
9386 S KRISTA DR E
GOODYEAR, AZ 85338
GRADING AND DRAINAGE PLAN

JOB NUMBER 98196		SHEET 2 OF 5	
DRAWING G&D			
DRAFTSMAN	CHECKED BY	DATE 04/17/2025	

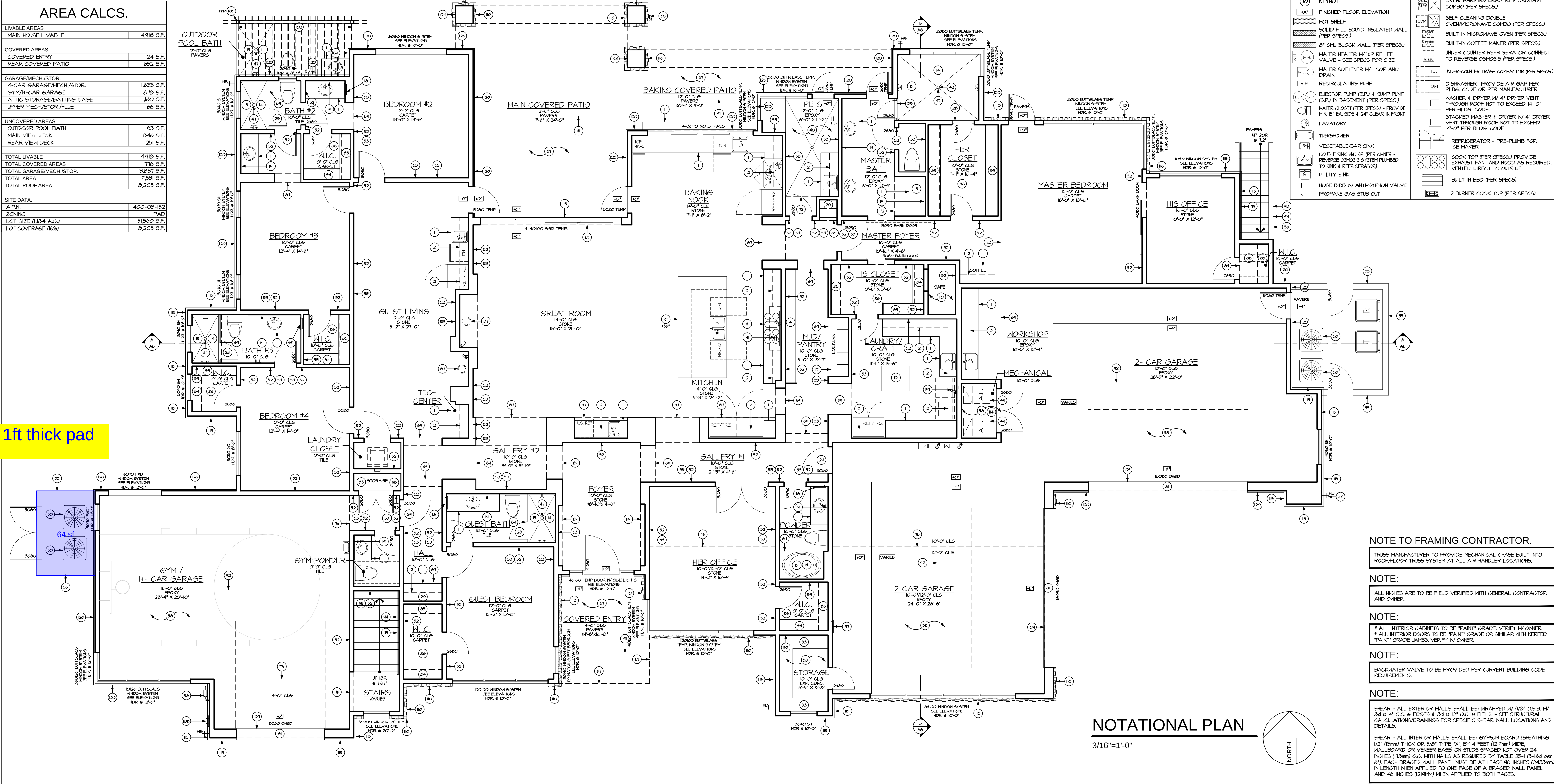
GENERAL INFORMATION	
1. WALL FRAMING - UNO. EXTERIOR WALLS - 2x6 @ 16" O.C. - UNO. INTERIOR BEARING WALLS - 2x4 @ 16" O.C. - UNO. INTERIOR NON BRG. - 2x4 @ 24" O.C. - UNO. 2. INSULATION: SPRAY FOAM INSULATION - CORR 1043 = TO R-39 ROOF - UNO. TO HAVE FIRE RETARDANT (1") = TO R-21 @ EXT. 2x6 WALLS - UNO. (6") = TO R-13 MIN. BTK. GARAGE & RESIDENCE (4") WALLS & CEILING - TYP. SOUND INSULATED WALLS SHALL BE SOLID FILL - TYP. 3. ALL WINDOW AND DOOR HEADER HEIGHTS TO BE 8'-0" - UNO. 4. SHOWER HEADS @ 78" AFF. UNO. SHOWER CONTROL VALVES @ 42" AFF. - UNO. STACK SHOWER CONTROL VALVES @ CURVED WALLS - UNO. 5. PROVIDE PRESSURE BALANCE OR THERMO. MIXING TYP. CONTROL VALVES FOR ALL SHOWER AND TUB COMBINATIONS. 6. GLASS BLOCK TO COMPLY WITH BUILDING CODE. 7. ALL BATH ACCESSORIES (TOWEL BARS, HOOKS ETC.) AND MOUNTING HEIGHTS TO BE DETERMINED BY OWNER. 8. PROVIDE BLOCKING IN WALLS AS NECESSARY TO SUPPORT ALL WALL MOUNTED FIXTURES. 9. ALL MECH. EQUIPMENT SHALL BE SCREENED A MINIMUM OF 12" ABOVE THE HIGHEST POINT OF THE EQUIPMENT. 10. ALL CEILING HEIGHTS INDICATED ARE FROM FINISHED FLOOR ELEVATION IN ADJACENT ROOM. 11. ALL LOW/PONY/BANCO WALL HEIGHTS TO BE MEASURED FROM HIGHEST ADJACENT FINISH FLOOR. 12. ALL INTERIOR AND EXTERIOR CORNERS TO HAVE 1/2" BULLNOSE EDGES UNLESS NOTED OTHERWISE. 13. ALL EXTERIOR DOORS EXITS FROM LIVABLE AREAS SHALL BE SOLID CORE, OR INSULATED WITH HEATHER TIGHT GASKETS AND THRESHOLDS OR GASKETED GLASS. 14. ALL EXTERIOR WINDOWS ADJACENT TO LIVABLE AREAS SHALL BE DUAL PANE. 15. LOCATION OF GAS METER AND BUILDING WATER MAIN SHUT-OFF PER UTILITY. 16. PROVIDE ROUGH GAS STUB FOR ALL STANDARD AND FUTURE GAS APPLIANCES PER GAS SCHEMATIC. 17. ALL EQUIPMENT IN GARAGE SHALL HAVE ELECTRIC (OR GAS) IGNITION POINTS AT 18" ABOVE FINISH FLOOR AND SHALL BE PROTECTED FROM DAMAGE. 18. XOX WINDOW = TO HAVE ONE OPERABLE WINDOW TO BE 5:1 5/8" MIN. WITH MIN. CLEAR DIM. OF 20" WIDE X 24" HIGH. 19. DUCTS FROM UNITS THRU CEILING SHALL BE CONSTRUCTED OF STEEL (MINIMUM 26 GAUGE) WITH NO OPENINGS WITHIN GARAGE. 20. NOISE ATTENUATION MEASURES ARE TO BE INCORPORATED INTO THE DESIGN AND CONSTRUCTION OF THE HOUSE TO ACHIEVE A NOISE LEVEL REDUCTION OF 25dB. PROVIDE THE MINIMUM FOLLOWING REQUIREMENTS: A. EXTERIOR WALLS - R-11 INSULATION B. CEILING - R-14 INSULATION C. EXTERIOR DOORS SHALL BE SOLID CORE WOOD OR INSULATED WITH GASKETS AND GASKETED GLASS D. ALL WINDOWS SHALL BE DOUBLE GLAZE E. SOLE PLATES OF EXTERIOR WALLS SHALL BE CAULKED OR SEALED AT THE FLOOR LINE. 21. EXTERIOR DOORS AND WINDOWS SHALL COMPLY WITH IRC. 22. PROVIDE AN APPROVED BARRIER BETWEEN FACTORY FIREPLACE BOXES AND CHIMNEYS, AND BUILDING INSULATION. CLEARANCE SHALL BE MAINTAINED BETWEEN THE CHIMNEY AND COMBUSTIBLE BARRIERS AND BETWEEN THE FIREPLACE OPENING AND ANY COMBUSTIBLE MATERIAL OUTSIDE THE FIREBOX PER MANUFACTURER'S INSTALLATION INSTRUCTIONS. 23. ANY DOORS USED FOR BEDROOM EGRESS SHALL NOT BE KEYED ON THE INTERIOR SIDE. 24. WATER HEATER DRAIN PAN SHALL EXTEND FULL-SIZE AND TERMINATE OVER A SUITABLE LOCATED INDIRECT MASTER RECEPTOR OR SHALL EXTEND TO THE EXTERIOR OF THE BLDG AND TERMINATE NOT LESS THAN 6 INCHES AND NOT MORE THAN 24 INCHES ABOVE THE ADJACENT GROUND SURFACE. 25. GYPSUM BOARD APPLIED TO A CEILING SHALL BE 1/2" WHEN FRAMING MEMBERS ARE 16" O.C. OR 5/8" WHEN FRAMING MEMBERS ARE 24" O.C. OR USE LABELED 1/2" SAG-RESISTANT GYPSUM CEILING BOARD. (TABLE R702.3.5 (6)). 26. PROVIDE A LISTED AND APPROVED RANGE AND/OR COOKTOP NOT INSTALLED IN ACCORDANCE WITH THE LISTING AND WITH THE MANUFACTURER'S INSTALLATION INSTRUCTIONS. VERIFY AND MAINTAIN REQUIRED HORIZONTAL AND VERTICAL CLEARANCES ABOVE THE FINISHED COUNTERTOP SURFACE BEFORE ORDERING OR INSTALLING CABINETS. 27. GLAZING MAXIMUM U-FACTOR 0.40, SOLAR HEAT GAIN COEFFICIENT (SHGC) MAXIMUM 0.25 (TABLE N102.1.1) 28. OPAQUE DOORS SEPARATING CONDITIONED AND UNCONDITIONED SPACE-MAXIMUM U-FACTOR 0.40. (N102.1.1). 29. SKYLIGHT - MAXIMUM U-FACTOR 0.65. (N102.1.1).	

ESTRELLA NOTES	
1. THE SE5 (ELECTRIC PANEL), HVAC (HEATING VENTILATING, AND AIR-CONDITIONING) UNITS, AND GAS METER INCLUDING THE REQUIRED SCREEN WALLS MUST BE LOCATED WITHIN THE BUILDING ENVELOPE. 2. EXTERIOR DOORS AND WINDOWS MUST BE RECESSED 4". NOTE: THIS CANNOT BE ACHIEVED WITH A 2X4 STUD WALL. 3. GARAGE DOOR MUST BE RECESSED 12". 4. SATELLITE DISHES MUST BE SCREENED TO THE EXTENT PRACTICABLE WHEN VIEWED FROM THE STREET. DISH MUST BE PAINTED TO MATCH THE ADJACENT HOME SURFACE. 5. CHANGES TO THE APPROVED PLANS AND ELEVATIONS THAT INVOLVE THE EXTERIOR DESIGN, MATERIALS, AND COLORS MUST BE SUBMITTED AND PROCESSED THROUGH THE DRG. SUBMIT CHANGES TWO WEEKS PRIOR TO THE MEETINGS TO AVOID A DELAY IN CONSTRUCTION.	

FLOOR PLAN KEYNOTES									
* SEE SPECIFICATION FOR MORE INFORMATION * ALL KEYNOTES MAY NOT APPLY TO THIS SHEET *									
No.	DESCRIPTION	No.	DESCRIPTION	No.	DESCRIPTION	No.	DESCRIPTION	No.	DESCRIPTION
KITCHEN	1 BASE CABINET W/COUNTERTOP- (PER SPECS.)	16 RECESSED MEDICINE CABINET (PER SPECS.)	31 SUPPLY A 14"x3" GRILLE ABOVE DOOR FOR UPPER COMBUSTION AIR	52 2x6 WALL	76 HEAT-N-GLO GAS F.P. 6000MOD-IFT BUILT TO CSA/ANSI Z21.80-14 & CSA 2.33-14 STANDARDS PER MFR. SPECS OR APPROVED EQUAL. DIRECT VENT RECESSED NICHIE. ADV. F.P. FOR TV	101 24 X 24 COLUMNS			
	2 UPPER CABINETS- (PER SPECS.)	17 48" MIR. ENTIRE LENGTH OF VANITY @ 3" ABOVE BACK SPLASH, PER SPECS	32 HIGH AND LOW COMBUSTION & RELIEF GRILLES (14"x20") TO BE LOCATED 12" FROM FLR. & CLG. IN GAS APPL.	53 SOLID FILL SOUND INSULATED WALL (PER SPECS.)	77 SOFFIT @ 9'-6" A.F.F.	102 4x4 POST (VERIFY W/ OWNER)			
	3 BREAKFAST BAR W/COUNTERTOP - (PER SPECS.) HEIGHT PER PLAN	18 LINEN. 5 ADJUSTABLE SHELVES EVENLY SPACED. (PER SPECS.)	33 GAS ENTRANCE. SEE GENERAL NOTES	54 LOW WALL, SEE PLAN FOR HEIGHT	78 SOFFIT @ 10'-0" A.F.F.	103 6x6 ROUGH-SAWN OR FAUX WOOD BEAM. BOTTOM OF BEAM @ 8" BELOW FINISHED CEILING. (PER OWNER)			
	4 PANTRY: 5 ADJUSTABLE SHELVES EVENLY SPACED. 4 SHELVES BELOW WINDOW (PER SPECS.)	19 LINEAR FLOOR DRAIN (PER SPECS)	34 PROVIDE 100 SQ. IN. COMBUSTION AIR FOR GAS DRYER	55 EXTERIOR LOW WALL - PAINT TO MATCH EXTERIOR OF HOUSE - SEE ELEVATIONS FOR MORE INFORMATION.	79 SOFFIT @ 11'-0" A.F.F.	104 6x6 ROUGH-SAWN OR FAUX WOOD BEAM. BOTTOM OF BEAM @ 6" BELOW FINISHED CEILING. (PER OWNER)			
	5 APPLIANCE GARAGE	20 RECESSED SHOWER ACCESSORIES NICHIE. (PER SPECS.)	35 FLOOR SINK FOR WASHING DOGS, PER SPECS. VERIFY W/ OWNER	56 SLOPED WALL @ 12" ABOVE NOSING OF TREAD	80 SOFFIT @ 12'-0" A.F.F.	105 2" OVERALL			
	6 BUILT-IN W/WOVEN (PER SPECS.)	21 RAISED PLATFORM FOR TUB. FINISH PER SPECS. HEIGHT PER PLAN	36 PLUMB FOR POT FILLER FAUCET	57 1/2" "GOLD BOND" EXTERIOR SPOFF BOARD (1-1/4" ER-1936) OR APPROVED EQUAL	81 SOFFIT @ 12'-6" A.F.F.	106 GUARDS TO BE 36" MIN. HEIGHT ABOVE WALKING SURFACE (PER CODE REQUIREMENTS). RAILS SHOULD BE SPACED TO NOT ALLOW A 4" SPHERE TO PASS THROUGH ANY OPENING. SEE ELEVATIONS FOR MORE INFO.			
	7 BUILT-IN WINE GLASS DISPLAY (PER SPECS.)	22 FRAMED SHOWER SEAT @ 24" A.F.F., FINISH MATERIAL. (PER SPECS.)	37 OVERHEAD SHOWER FIXTURES/LETS. (PER OWNER & SPECIFICATIONS)	58 5/8" TYPE "X" GYP. BD. @ GARAGE CLG. WHEN GARAGE IS UNDER A HABITABLE ROOM, USE 1/2" GYP. BOARD @ ALL OTHER WALLS & CLG PER IRC R302.6	82 22"x30" ATTIC ACCESS (PER SPECS.)	107 ELECTRIC PANEL LOCATION-SEE ELEC. PLAN AND GENERAL NOTES			
	8 HOOD OVER BBQ. VENT TO EXTERIOR MAX 400 C.F.M. UNO.	23 TOWEL HOOK (PER SPECS)	38 FRAMED PLATFORM RAISED 18" MIN. ABOVE FINISH FLOOR	59 1/2" TYPE "X" GYP. BD. @ GARAGE CLG. WHEN GARAGE IS UNDER A HABITABLE ROOM, USE 1/2" GYP. BOARD @ ALL OTHER WALLS & CLG PER IRC R302.6	83 EXACT LOCATION - DESIGN TRUSSES ACCORDINGLY. REFER TO DETAIL ON SHEET A6.1 FOR MORE INFORMATION.	108 GARAGE CONC. FLOOR TO STOP AT INSIDE WALL. 50' DRIVEWAY FINISH MATERIAL CAN BE APPLIED HERE			
	9 KITCHEN ISLAND W/COUNTERTOP - (PER SPECS.) HEIGHT PER PLAN	24 RECESS SLAB @ SHOWER TO PROVIDE CURB LESS FLUSH ENTRY	39 BUILDING WATER MAIN SHUT-OFF VALVE LOCATION	60 NICHIE: SILL @ 3'-0" TOP @ 10'-0"	84 AUD/VIS. NICHIE - VERIFY W/ OWNER	109 ELDDORADO STONE, LLC. VENEER PER ICC REPORT #ESR-1125. VERIFY FOR EQUAL. SEE SPECS FOR MORE INFORMATION.			
	BATH	1 NET BAR W/COUNTERTOP - (PER SPECS.) HEIGHT PER PLAN	25 FULL HEIGHT MIRROR (PER SPECS)	40 TEMPERATURE AND PRESSURE RELIEF. MIN. 6" ABOVE GRADE AND 36" AWAY FROM OPENINGS	61 NICHIE: SILL @ 3'-0" TOP @ 10'-0"	85 BRISTOL DAYLIGHT SYSTEMS SKYLIGHT MODEL 2246 (ICC #ESR-311) OR SIMILAR	110 CONCRETE OR MASONRY BASEMENT WALL (SEE STRUCT. DRAWINGS)		
2 BUILT IN TABLE - (PER SPECS.) HEIGHT PER PLAN		26 TEMPERED GLASS DOOR SHOWER ENCLOSURE (PER SPECS.)	41 PLUMB TO SOFT WATER LOOP	62 ARCHED NICHIE: SILL @ 3'-0" TOP @ 8'-0" - FIELD VERIFY	86 BUILT IN STORAGE CABINETS - PER OWNER	111 WALK DECK SURFACE BY POLYCOAT PRODUCTS PER ICC #ESR-2185 AND SPECIFICATIONS - OR APPROVED EQUAL.			
3 MAKE-UP/VANITY HEIGHT PER PLAN (PER SPECS.)		27 1-3/8" THICK SOLID CORE DOOR, TIGHT FITTING, SELF-CLOSING & SELF-LATCHING	42 RECESS SLAB 1" AND PROVIDE DRAIN	63 ARCHED NICHIE: SILL @ 3'-0" TOP @ 8'-0" - FIELD VERIFY	87 EXTERIOR SOFFIT - SEE ELEV. FOR MORE INFORMATION	112 RECESS DOOR TRAIL IN CONCRETE PER SPECS. VERIFY W/ OWNER			
4 CEMENT, FIBER-CEMENT OR GLASS MAT GYPSUM BACKERS SHALL BE USED AS BACKERS FOR WALL TILE IN TUB AND SHOWER AREAS, PER IRC R102.4.2		28 ACTIVE DOOR	43 MECHANICAL CHASE	64 ARCHED NICHIE: SILL @ 3'-0" TOP @ 11'-0" - LESS 6" - FIELD VERIFY	88 BUILT IN DESK - PER OWNER	113 PREBENDING NOT TUB (PER SPECS.)			
5 TUB & WALL COVERS (PER SPECS.) OR WALLS W/ TILE TO MIN. 12"		29 LOUVERED MECH. DOORS	44 AIR HANDLER UNIT - SEE MECH. PLAN FOR MORE INFO. NOTE TO TRUSS MANUFACTURER: PROVIDE FOR MECHANICAL CHASE IN TRUSS DESIGN	65 ARCHED NICHIE: SILL @ 3'-0" TOP @ 11'-0" - LESS 6" - FIELD VERIFY	89 5 SHELVES EVENLY SPACED.	114 GAS STUB OUT FOR POOL/SPA EQUIP. SEE SITE FOR LOCATION			
6 LETTID (W/ ELECTRIC DISCONNECT) OR SOAKING TUB (PER SPECS.)		30 PET DOOR (PER SPECS)	45 A/C CONDENSING UNIT - SEE MECH. PLAN FOR MORE INFO. HOLD 6" AWAY FROM HOUSE	66 TOP OF ARCH @ 10'-0", LESS 6" - FIELD VERIFY	90 1 ROD, 1 SHELF	115 GAS STUB OUT FOR BBQ. & FIREPIET. SEE SITE FOR LOCATION			
7 ACCESS PANEL FOR MOTOR & HEATER @ TUB UNDER COUNTER.		31 DECORATIVE WROUGHT IRON AND/OR WOOD GATE - SEE ELEVATIONS	46 8" CMU/STUCCOED SCREEN WALL MIN. 12" ABOVE MECH. EQUIP.	67 TOP OF ARCH @ 11'-0", LESS 6" - FIELD VERIFY	91 2 RODS, 2 SHELVES	116 CONCENTRIC VENTS TO EXTERIOR			
		32 FACE FRAME CABINET DOORS. VERIFY SIZE AND LOCATION WITH OWNER		68 TOP OF ARCH @ 12'-0", LESS 6" - FIELD VERIFY	92 CLOSET SHOELVES BY CLOSET SUBCONTRACTOR AND SPECS.	117 METAL PANELS (PER SPECS.)			
		33 DECORATIVE WROUGHT IRON FRONT ENTRY DOOR (PER SPECS.) SEE ELEV. FOR MORE INFO.		69 SOFFIT @ 13'-0" A.F.F.	93 HEAT-N-GLO GAS F.P. 6000MOD-IFT BUILT TO CSA/ANSI Z21.80-14 & CSA 2.33-14 STANDARDS PER MFR. SPECS OR APPROVED EQUAL.	118 BUILT-IN BENCH (PER SPECS.)			
				70 SOFFIT @ 8'-6" A.F.F.	94 LINE ABOVE OR BELOW	119 BUILT-IN LOCKERS (PER SPECS.)			
			71 SOFFIT @ 9'-0" A.F.F.		120 BUILT-IN SAFE (PER SPECS.)				
					121 BUILT-IN CRAFT ISLAND (PER SPECS.)				
					122 WOOD TILE (PER SPECS.)				

S Y M B O L L E G E N D

AREA CALCS.	
LIVABLE AREAS	
MAIN HOUSE LIVABLE	4,918 S.F.
COVERED AREAS	
COVERED ENTRY	124 S.F.
REAR COVERED PATIO	652 S.F.
GARAGE/MECH./STOR.	
4-CAR GARAGE/MECH./STOR.	1,638 S.F.
GYM/1-CAR GARAGE	878 S.F.
ATTIC STORAGE/BATTING CAGE	1,600 S.F.
UPPER MECH./STOR./FLUE	166 S.F.
UNCOVERED AREAS	
OUTDOOR POOL BATH	83 S.F.
MAIN VIEW DECK	846 S.F.
REAR VIEW DECK	251 S.F.
TOTAL LIVABLE	4,918 S.F.
TOTAL COVERED AREAS	716 S.F.
TOTAL GARAGE/MECH./STOR.	3,237 S.F.
TOTAL AREA	4,531 S.F.
TOTAL ROOF AREA	8,205 S.F.
SITE DATA:	
A.P.N.	400-03-152
ZONING	PAD
LOT SIZE (1.04 A.C.)	51,560 S.F.
LOT COVERAGE (16%)	8,205 S.F.



NOTE TO FRAMING CONTRACTOR:	
TRUSS MANUFACTURER TO PROVIDE MECHANICAL CHASE BUILT INTO ROOF/FLOOR TRUSS SYSTEM AT ALL AIR HANDLER LOCATIONS.	
NOTE:	
ALL NICHES ARE TO BE FIELD VERIFIED WITH GENERAL CONTRACTOR AND OWNER.	
NOTE:	
* ALL INTERIOR CABINETS TO BE "PAINT" GRADE. VERIFY W/ OWNER. * ALL INTERIOR DOORS TO BE "PAINT" GRADE OR SIMILAR WITH KERPED "PAINT" GRADE JAMBS. VERIFY W/ OWNER.	
NOTE:	
BACKHATCH VALVE TO BE PROVIDED PER CURRENT BUILDING CODE REQUIREMENTS.	
NOTE:	
SHEAR - ALL INTERIOR WALLS SHALL BE WRAPPED W/ 3/8" O.S.B. W/ 8d @ 4" O.C. @ EDGES & 8d @ 12" O.C. @ FIELD. - SEE STRUCTURAL CALCULATIONS/DRAWINGS FOR SPECIFIC SHEAR WALL LOCATIONS AND DETAILS. SHEAR - ALL INTERIOR WALLS SHALL BE, GYPSUM BOARD (SHEATHING 1/2" (3mm) THICK OR 5/8" TYPE 'X', BY 4 FEET (1219mm) WIDE, WALLBOARD OR VENEER BASED ON STUDS SPACED NOT OVER 24 INCHES (178mm) O.C. WITH NAILS AS REQUIRED BY TABLE 25-1 (3-16d @ 6") EACH BRACED WALL PANEL MUST BE AT LEAST 16 INCHES (406mm) IN LENGTH WHEN APPLIED TO ONE FACE OF A BRACED WALL PANEL AND 48 INCHES (1219mm) WHEN APPLIED TO BOTH FACES.	

Koch-Haut Residence

A.P.N. 400-03-152

Lot 79 of Estrella Parcel 9

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www.iplandesign.com

Home Builders Association

OF CENTRAL ARIZONA

INSTITUTE OF BUILDING DESIGN

A B D.

PROFESSIONAL BUILDING DESIGNER

MAIN LEVEL

NOTATIONAL FLOOR PLAN

PLOT DATE:

Rev. 1 DATE:

2

3

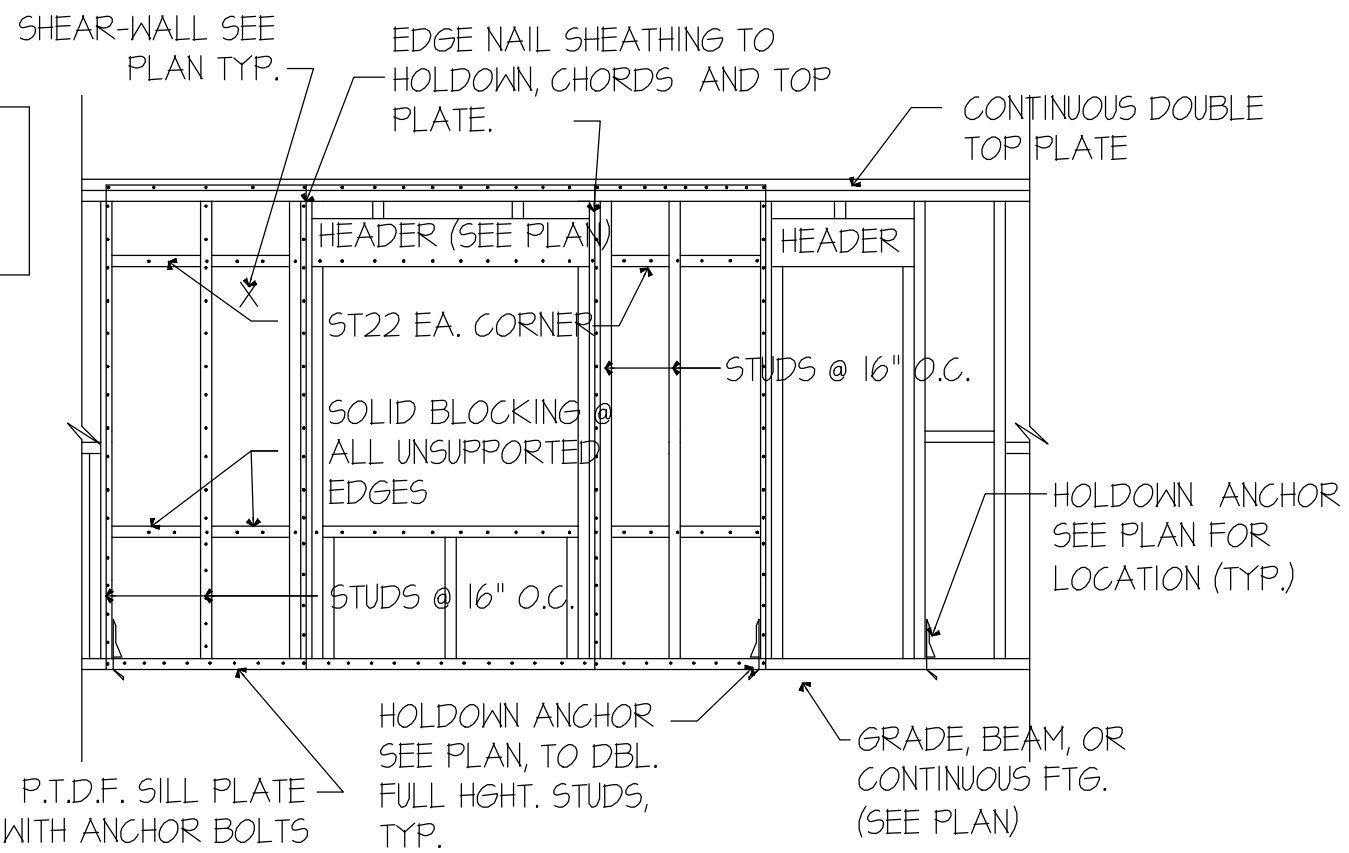
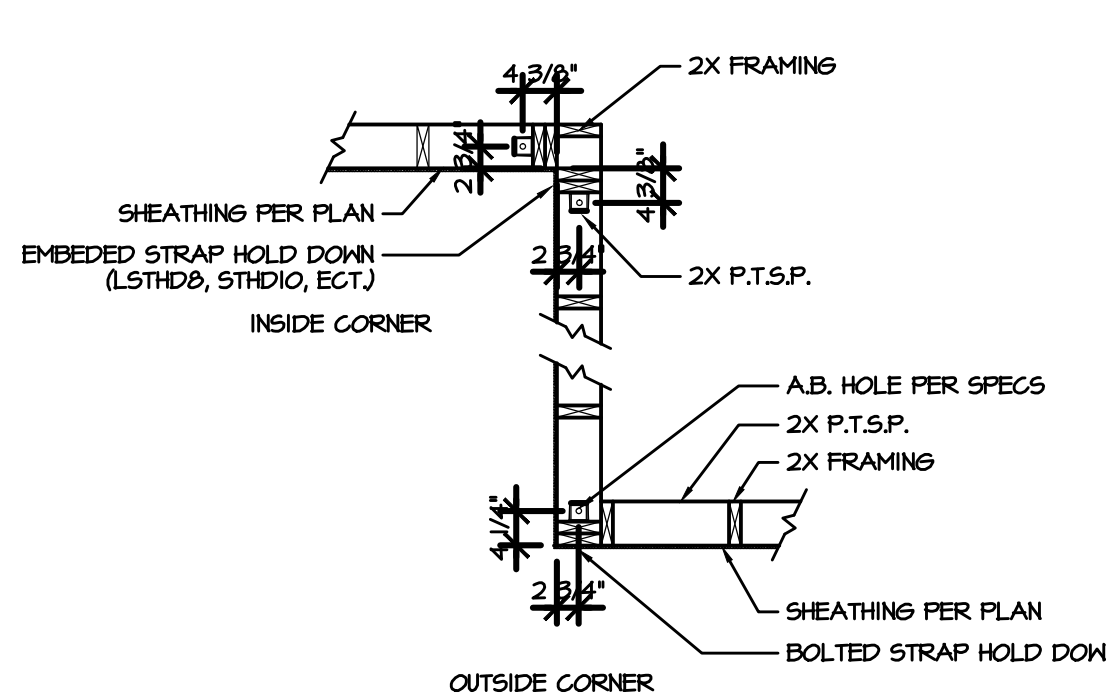
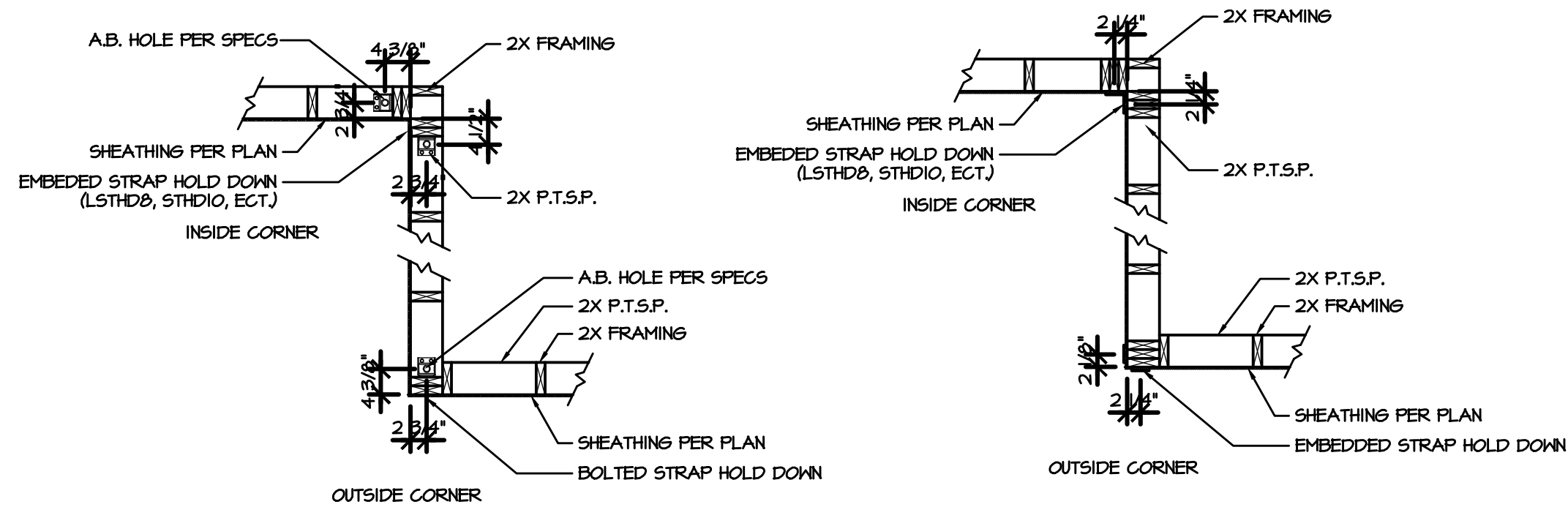
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PLAN #

7467

SHEET NO.

A2

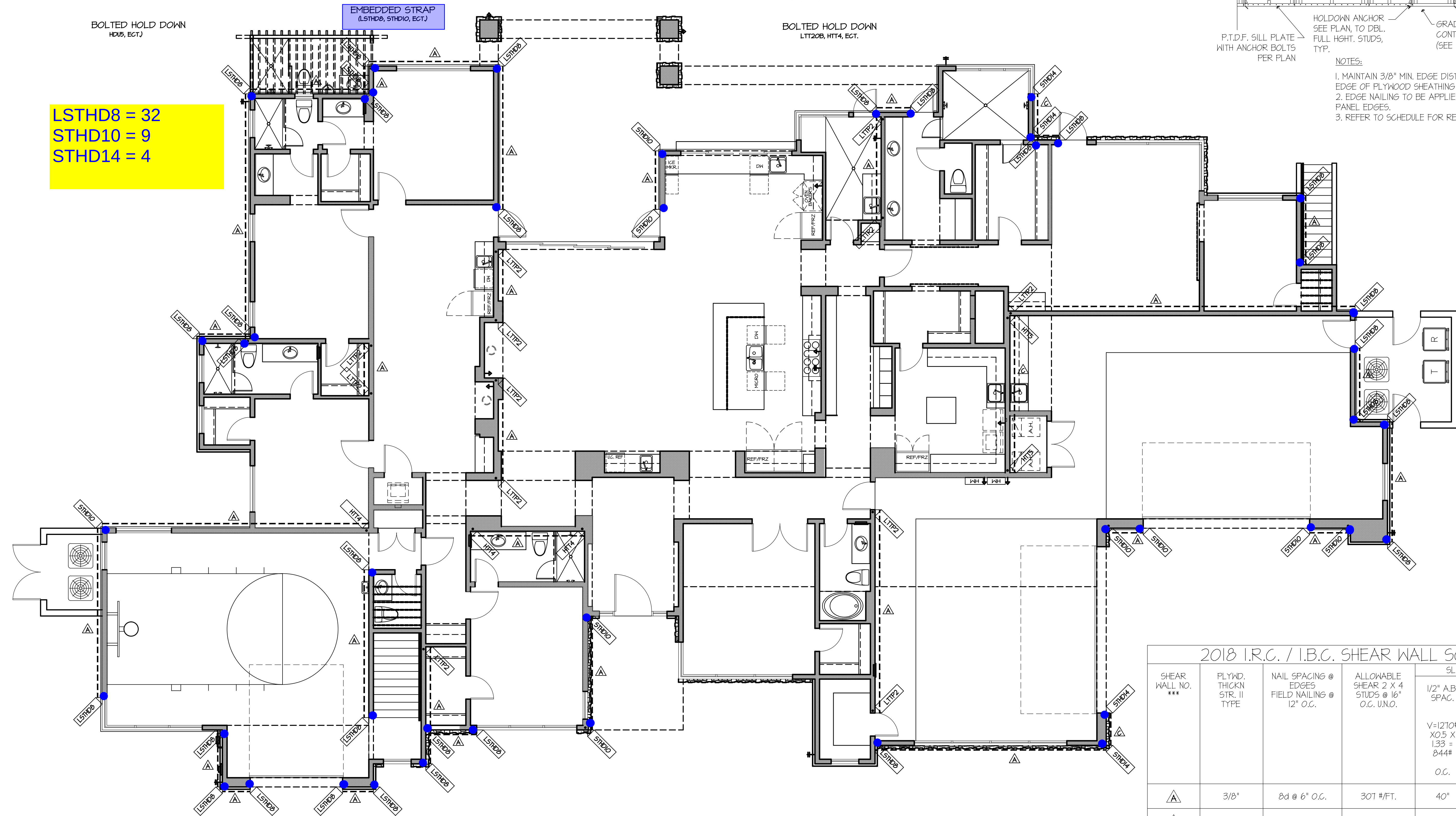


FOR ALL GENERAL NOTES SEE SHEET G5N

NOTE:
SHEAR WHOLE HOUSE PER SHEAR WALL SCHEDULE "A" U.N.O.

- NOTES:
1. MAINTAIN 3/8" MIN. EDGE DISTANCE FROM EDGE OF PLYWOOD SHEATHING TO NAILS.
 2. EDGE NAILING TO BE APPLIED TO ALL PANEL EDGES.
 3. REFER TO SCHEDULE FOR REQUIRED NAILING.

LSTHD8 = 32
STHD10 = 9
STHD14 = 4



2018 I.R.C. / I.B.C. SHEAR WALL SCHEDULE						
SHEAR WALL NO. ***	PLYND. THICKN STR. II TYPE	NAIL SPACING @ EDGES FIELD NAILING @ 12" O.C.	ALLOWABLE SHEAR 2 X 4 STUDS @ 16" O.C. U.N.O.	SLIDING ANCHOR SYSTEM		
				1/2" A.B. SPAC.	A35 OR EQUIV.	16d COMMON NAIL SPAC.
A	3/8"	8d @ 6" O.C.	307 #/FT.	40"	24"	6"
B	3/8"	8d @ 4" O.C.	447 #/FT.	24"	16"	4"
C	3/8"	8d @ 3" O.C.	575 #/FT.	18"	12"	3"
D*	3/8"	8d @ 2" O.C.*	743 #/FT.	16"	10"	2"

* 3X NOMINAL FRAMING
** ALL JOINTS SHALL BE BLOCKED.
*** WHEN SPECIFIED ON BOTH SIDES, CUT SLIDING ANCHOR SYSTEM CONNECTIONS IN HALF.
**** ALL RETROFITED HOLD DOWNS TO BE INSTALLED WITH EPOXYED ALLTHREAD PER ENGINEER.

4128 N. 34TH STREET
PHOENIX, AZ 85018
PHONE: (480) 874-7190
FAX: (480) 874-7185

Palos Verdes Engineering Corporation

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Koch-Haut Res.
9386 S. Krista Dr.
Goodyear, AZ 85338

REGISTERED PROFESSIONAL ENGINEER
SHEETS
WILLIAM J. CHRISTENSEN
5/10/2025
GOODYEAR, ARIZONA, U.S.A.

REVISIONS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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DATE: _____
DRAWN BY: _____
JOB NO.: _____
REVISION NO.: _____

SHEET NO.
S-1
OF

PAD SCHEDULE		
PAD	SIZE	REINFORCEMENT
(F1)	2'-0" X 2'-0" X 12"	2 - #5 EA. WAY
(F2)	2'-6" X 2'-6" X 12"	3 - #5 EA. WAY
(F3)	3'-0" X 3'-0" X 12"	3 - #5 EA. WAY
(F4)	3'-6" X 3'-6" X 12"	3 - #5 EA. WAY
(F5)	4'-0" X 4'-0" X 12"	4 - #5 EA. WAY
(F6)	4'-6" X 4'-6" X 12"	4 - #5 EA. WAY
(F7)	6'-0" X 6'-0" X 24"	5 - #5 EA. WAY

FOR HOLD DOWNS WITH $\frac{5}{8}$ " DIA. BOLT - USE 5B5/8X24 OR 55TB24 - FOOTING TO BE 30" DEEP.

FOR HOLD DOWNS WITH $\frac{3}{4}$ " DIA. BOLT - USE 5B7/8X24 OR 55TB28 - FOOTING TO BE 30" DEEP.

FOR HOLD DOWNS WITH 1" DIA. BOLT - USE 5B1X30 - FOOTINGS TO BE 36" DEEP.

KEYNOTES

* SEE SPECIFICATION FOR MORE INFORMATION *

* ALL KEYNOTES MAY NOT APPLY TO THIS SHEET *

No.	DESCRIPTION	No.	DESCRIPTION
①	4" CONCRETE SLAB OVER 4" ABC OVER TERMITE TREATED SOIL.	⑥	RECESSED ELECTRICAL - VERIFY LOCATION WITH OWNER
②	6" STEM WALL WITH #4 HORIZ. AT TOP AND #4 # 48" O.C. VERT. 48 BAR DIA. OVERLAP.	⑦	12" STEM WALL WITH #4 HORIZ. AT TOP AND #4 # 48" O.C. VERT. 48 BAR DIA. OVERLAP.
③	16" WIDE X 12" THICK FOOTING WITH 4-#4 BARS AT BOTTOM OF FOOTING. BOTTOM OF FOOTING 18" BELOW NATIVE SOIL OR ENGINEERED PAD.	⑧	22" WIDE X 10" THICK FOOTING WITH 2-#4 BARS AT BOTTOM OF FOOTING. BOTTOM OF FOOTING 18" BELOW NATIVE SOIL OR ENGINEERED PAD.
④	8" TURNDOWN - SEE DETAILS	⑨	VERIFY WITH CONTRACTOR FOR TRACK DEPTH AT SLAB
⑤	RECESSED SLAB FOR SHOWER DRAIN - SEE DETAILS	⑩	24" #4 BARS AT 32" O.C. - EPOXY WITH SIMPSON SET-XP EPOXY

FOR ALL GENERAL NOTES
SEE SHEET 65N

BOLT LEGEND

- A 1/2" DIA. A.B. X 10" @ 36" O.C., 7" EMB. # 3" SQ WASHER
- B 1/2" DIA. A.B. X 10" @ 24" O.C., 7" EMB. # 3" SQ WASHER
- C 1/2" DIA. A.B. X 10" @ 18" O.C., 7" EMB. # 3" SQ WASHER
- D 1/2" DIA. A.B. X 10" @ 12" O.C., 7" EMB. # 3" SQ WASHER

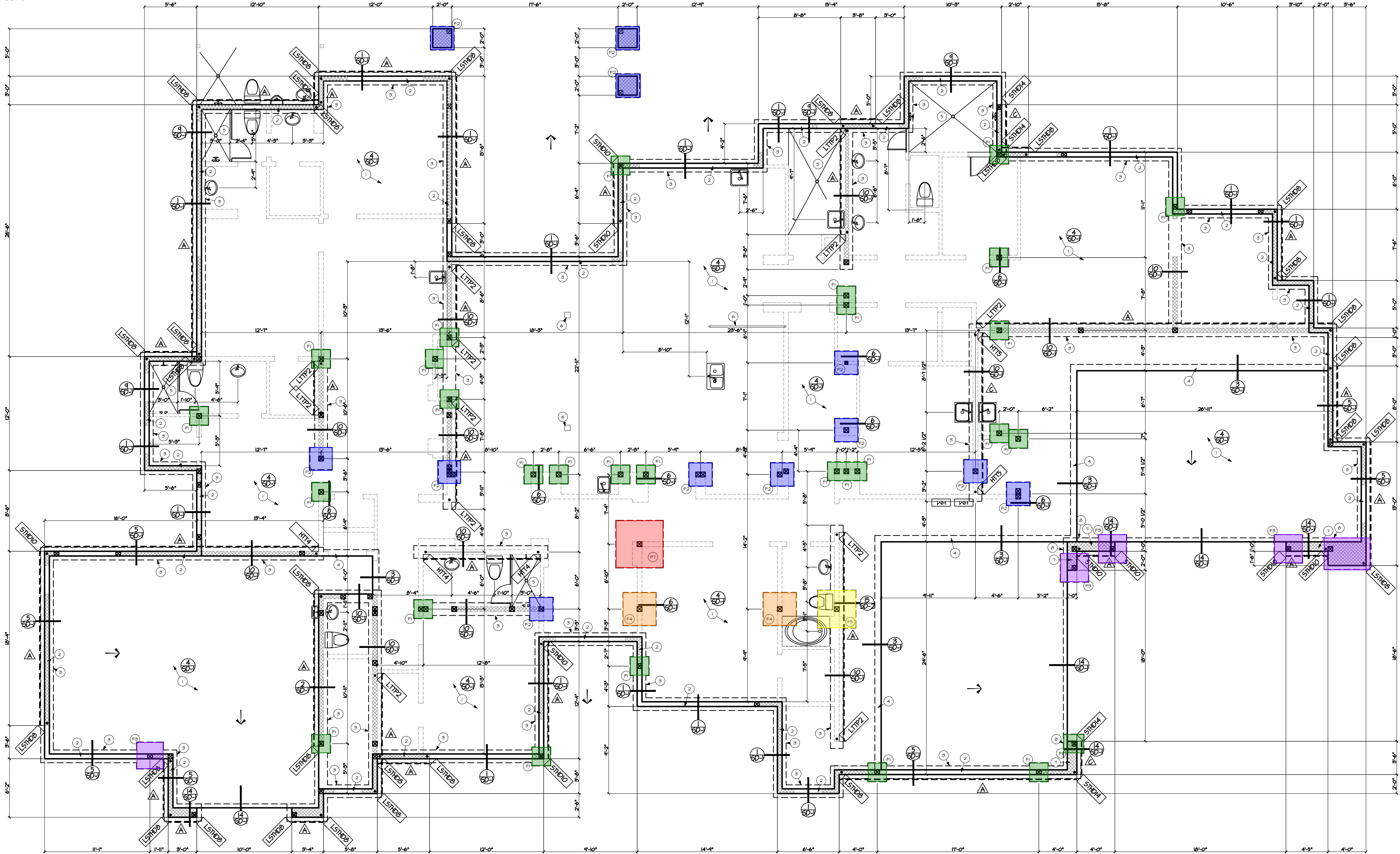
USE 1/2" DIA. A.B. X 10" @ 48" O.C.
UNLESS NOTED OTHERWISE

FOUNDATION NOTES

1. FINISH FLOOR SHALL BE A MINIMUM OF 8" ABOVE ADJACENT FINISH GRADE.
2. PROVIDE EXTERIOR WATERPROOF MEMBRANE UP TO 1'-0" BEYOND CONC. STEM AT ALL SUNKEN FLOORS.
3. SEAL ALL VOIDS AROUND PIPING PASSING THRU CONCRETE FLOORS.
4. ALL EXTERIOR CONCRETE FLATWORK TO SLOPE A MINIMUM OF 1/8". FT. AND ALL LANDINGS AT DOORS TO SLOPE A MAXIMUM OF 1/4". FT.
5. ALL INTERIOR AND EXTERIOR FOOTINGS SHALL BEAR 18" BELOW ENGINEER CERTIFIED COMPACTED FILL OR UNDISTURBED NATIVE SOIL WITH MINIMUM ALLOWABLE SOIL BEARING PRESSURE AT 1,000 P.S.F.
6. MIN. 28 DAY COMPREHENSIVE STRENGTH FOR CONCRETE SHALL BE 2500 P.S.I.
7. ALL PROCEDURES, PLACEMENT FORMWORK LAP, ETC. SHALL CONFORM WITH LATEST A.C.I. STANDARDS, MECHANICALLY VIBRATE ALL CONCRETE WHEN PLACED, EXCEPT THE SLABS ON GRADE NEED ONLY TO BE VIBRATED AROUND UNDERFLOOR DUCTS, ETC. MIN. SLUMP 4.5". ALL CONCRETE SLABS ON GRADE SHALL BE BOUNDED BY CONSTRUCTION JOINTS, KEYED OR SAWCUT SUCH THAT THE ENCLOSED AREA DOES NOT EXCEED 600 SQUARE FEET. ALL WORK IS TO CONFORM TO ALL LOCAL BUILDING CODES.
8. CONTINUOUS FLOOR SLAB SHALL BE SAWCUT EVERY 600 SQUARE FEET FOR EXPANSION. SAWCUTS SHALL BE UNDER INTERIOR NON-BEARING WALLS OR IN AREAS NOT AFFECTING TILE FLOORS WHERE POSSIBLE.
9. TOP OF EXTERIOR FOUNDATION SHALL BE 12" ±2% ABOVE THE ELEVATION OF THE STREET GUTTER OR INLET OF AN APPROVED DRAINAGE DEVICE.
10. ALL BOTTOM PLATES SHALL BE ANCHORED TO THE FOUNDATION WITH 1/2"-ANCHOR BOLTS HAVING 7" MIN. EMBEDMENT @ 4' O.C. THERE SHALL BE A MINIMUM OF TWO BOLTS PER PLATE SECTION WITH ONE BOLT LOCATED NOT MORE THAN 12" OR LESS THAN SEVEN BOLT DIAMETERS FROM EACH END OF THE PLATE SECTION.

iso footgin

f1 = 29
f2 = 12
f3 = 6
f4 = 2
f5 = 1
f6 = 0
f7 = 1



FOUNDATION PLAN

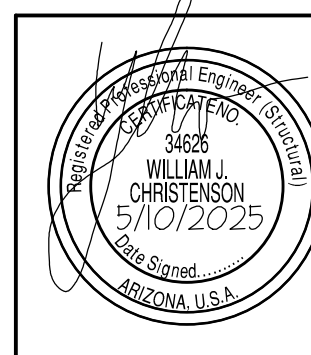
SCALE 3/16" = 1'-0"

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Koch-Haut Res.
9386 S. Krista Dr.
Goodyear, AZ 85338



REVISIONS	

DATE:	DRAWN BY:	JOB NO.:	REVISION NO.:

SHEET NO.

5-2
OF

PAD SCHEDULE		
PAD	SIZE	REINFORCEMENT
(F1)	2'-0" X 2'-0" X 12"	2 - #5 EA. WAY
(F2)	2'-6" X 2'-6" X 12"	3 - #5 EA. WAY
(F3)	3'-0" X 3'-0" X 12"	3 - #5 EA. WAY
(F4)	3'-6" X 3'-6" X 12"	3 - #5 EA. WAY
(F5)	4'-0" X 4'-0" X 12"	4 - #5 EA. WAY
(F6)	4'-6" X 4'-6" X 12"	4 - #5 EA. WAY
(F7)	6'-0" X 6'-0" X 24"	5 - #5 EA. WAY

FOR HOLD DOWNS WITH $\frac{5}{8}$ " DIA. BOLT - USE 5B5/8X24 OR 55TB24 - FOOTING TO BE 30" DEEP.

FOR HOLD DOWNS WITH $\frac{7}{8}$ " DIA. BOLT - USE 5B7/8X24 OR 55TB28 - FOOTING TO BE 30" DEEP.

FOR HOLD DOWNS WITH 1" DIA. BOLT - USE 5B1X30 - FOOTINGS TO BE 36" DEEP.

cont footing

16" x 12" = 650ft

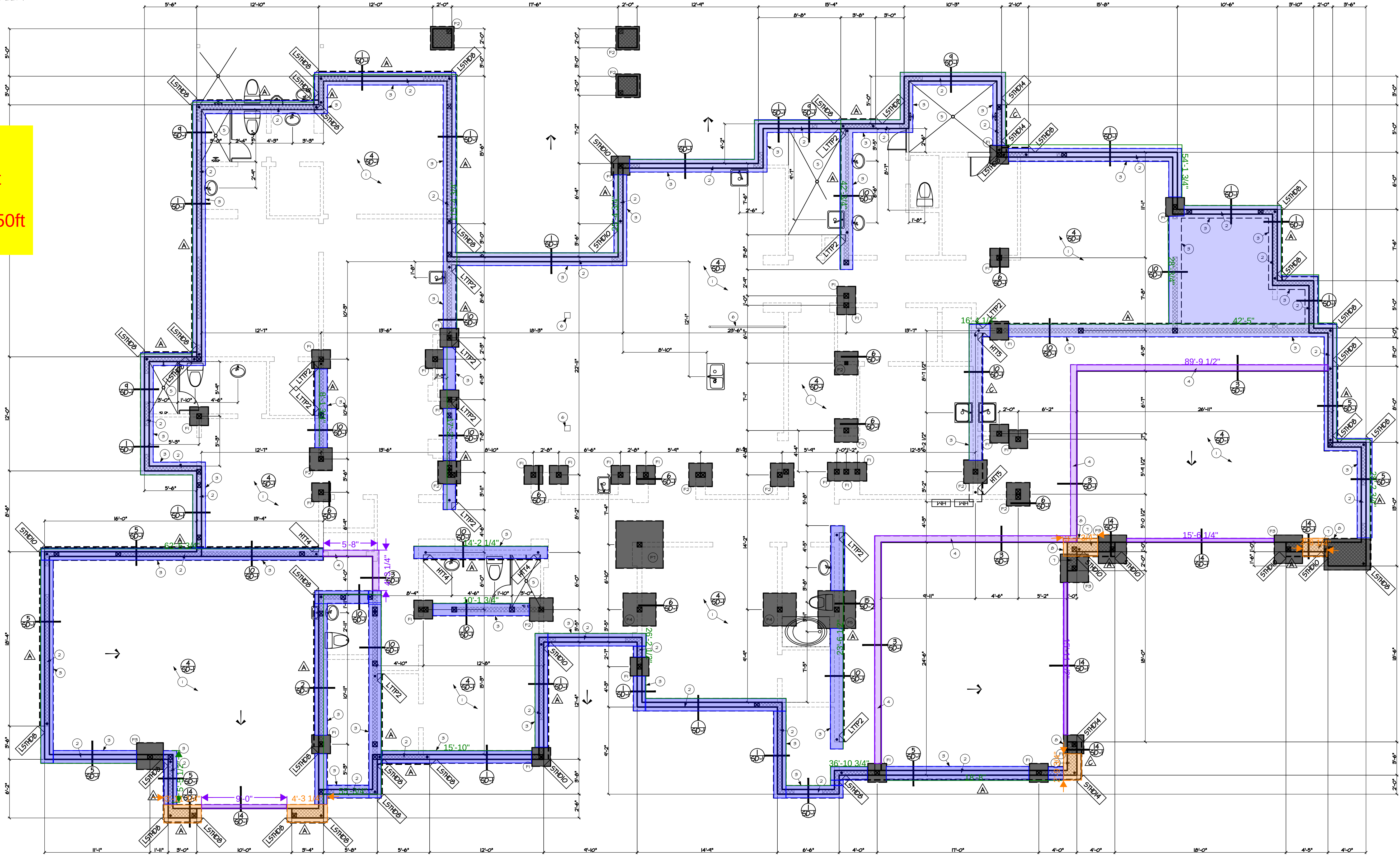
22" x 12" = 20ft

8" turndown = 150ft

KEYNOTES			
* SEE SPECIFICATION FOR MORE INFORMATION *			
* ALL KEYNOTES MAY NOT APPLY TO THIS SHEET *			
No.	DESCRIPTION	No.	DESCRIPTION
1	4" CONCRETE SLAB OVER 4" ABC OVER TERMITES TREATED SOIL.	6	RECESSED ELECTRICAL - VERIFY LOCATION WITH OWNER
2	6" STEM WALL WITH #4 HORIZ. AT TOP AND #4 # 48" O.C. VERT. 48 BAR DIA. OVERLAP.	7	12" STEM WALL WITH #4 HORIZ. AT TOP AND #4 # 48" O.C. VERT. 48 BAR DIA. OVERLAP.
3	16" WIDE X 12" THICK FOOTING WITH 4-#4 BARS AT BOTTOM OF FOOTING. BOTTOM OF FOOTING 18" BELOW NATIVE SOIL OR ENGINEERED PAD.	8	22" WIDE X 10" THICK FOOTING WITH 2-#4 BARS AT BOTTOM OF FOOTING. BOTTOM OF FOOTING 18" BELOW NATIVE SOIL OR ENGINEERED PAD.
4	8" TURNDOWN - SEE DETAILS	9	VERIFY WITH CONTRACTOR FOR TRACK DEPTH AT SLAB
5	RECESSED SLAB FOR SHOWER DRAIN - SEE DETAILS	10	24" #4 BARS AT 32" O.C. - EPOXY WITH SIMPSON SET-XP EPOXY

FOR ALL GENERAL NOTES SEE SHEET 65N	
BOLT LEGEND	
A	1/2" DIA. A.B. X 10" @ 36" O.C., 7" EMB. # 3" SQ WASHER
B	1/2" DIA. A.B. X 10" @ 24" O.C., 7" EMB. # 3" SQ WASHER
C	1/2" DIA. A.B. X 10" @ 18" O.C., 7" EMB. # 3" SQ WASHER
D	1/2" DIA. A.B. X 10" @ 12" O.C., 7" EMB. # 3" SQ WASHER
USE 1/2" DIA. A.B. X 10" @ 48" O.C. UNLESS NOTED OTHERWISE	

FOUNDATION NOTES	
1. FINISH FLOOR SHALL BE A MINIMUM OF 8" ABOVE ADJACENT FINISH GRADE.	7. ALL PROCEDURES, PLACEMENT FORMWORK LAP, ETC. SHALL CONFORM WITH LATEST A.C.I. STANDARDS, MECHANICALLY VIBRATE ALL CONCRETE WHEN PLACED, EXCEPT THE SLABS ON GRADE NEED ONLY TO BE VIBRATED AROUND UNDERFLOOR DUCTS, ETC. MIN. SLUMP 4.5". ALL CONCRETE SLABS ON GRADE SHALL BE BOUNDED BY CONSTRUCTION JOINTS, KEYED OR SAWCUT SUCH THAT THE ENCLOSED AREA DOES NOT EXCEED 600 SQUARE FEET. ALL WORK IS TO CONFORM TO ALL LOCAL BUILDING CODES.
2. PROVIDE EXTERIOR WATERPROOF MEMBRANE UP TO 1'-0" BEYOND CONC. STEM AT ALL SUNKEN FLOORS.	8. CONTINUOUS FLOOR SLAB SHALL BE SAWCUT EVERY 600 SQUARE FEET FOR EXPANSION. SAWCUTS SHALL BE UNDER INTERIOR NON-BEARING WALLS OR IN AREAS NOT AFFECTING TILE FLOORS WHERE POSSIBLE.
3. SEAL ALL VOIDS AROUND PIPING PASSING THRU CONCRETE FLOORS.	9. TOP OF EXTERIOR FOUNDATION SHALL BE 12" ± 2% ABOVE THE ELEVATION OF THE STREET GUTTER OR INLET OF AN APPROVED DRAINAGE DEVICE.
4. ALL EXTERIOR CONCRETE FLATWORK TO SLOPE A MINIMUM OF 1/8" FT. AND ALL LANDINGS AT DOORS TO SLOPE A MAXIMUM OF 1/4" FT.	10. ALL BOTTOM PLATES SHALL BE ANCHORED TO THE FOUNDATION WITH 1/2" ANCHOR BOLTS HAVING 7" MIN. EMBEDMENT @ 4' O.C. THERE SHALL BE A MINIMUM OF TWO BOLTS PER PLATE SECTION WITH ONE BOLT LOCATED NOT MORE THAN 12" OR LESS THAN SEVEN BOLT DIAMETERS FROM EACH END OF THE PLATE SECTION.
5. ALL INTERIOR AND EXTERIOR FOOTINGS SHALL BEAR 18" BELOW ENGINEER CERTIFIED COMPACTED FILL OR UNDISTURBED NATIVE SOIL WITH MINIMUM ALLOWABLE SOIL BEARING PRESSURE AT 1,000 P.S.F.	
6. MIN. 28 DAY COMPREHENSIVE STRENGTH FOR CONCRETE SHALL BE 2500 P.S.I.	



FOUNDATION PLAN

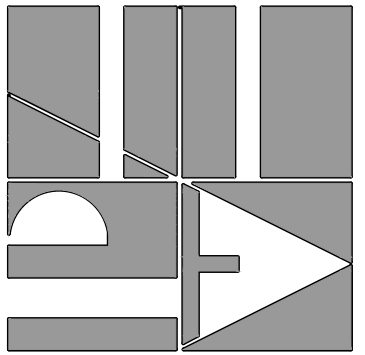
SCALE 3/16" = 1'-0"

PLTDATE:

CADFILE:

Palos Verdes
Engineering
Corporation

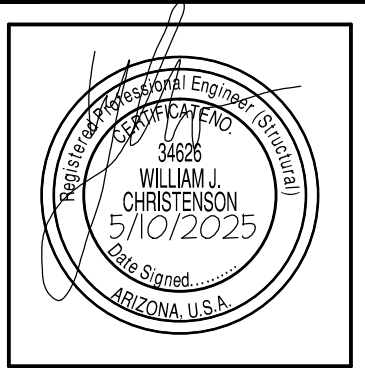
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Koch-Haut Res.

9386 S. Krista Dr.
Goodyear, AZ 85338



REVISIONS			
DATE:	DRAWN BY:	JOB NO.:	REVISION NO.:

SHEET NO.

5-2
OF

PAD SCHEDULE		
PAD	SIZE	REINFORCEMENT
(F1)	2'-0" X 2'-0" X 12"	2 - #5 EA. WAY
(F2)	2'-6" X 2'-6" X 12"	3 - #5 EA. WAY
(F3)	3'-0" X 3'-0" X 12"	3 - #5 EA. WAY
(F4)	3'-6" X 3'-6" X 12"	3 - #5 EA. WAY
(F5)	4'-0" X 4'-0" X 12"	4 - #5 EA. WAY
(F6)	4'-6" X 4'-6" X 12"	4 - #5 EA. WAY
(F7)	6'-0" X 6'-0" X 24"	5 - #5 EA. WAY

FOR HOLD DOWNS WITH $\frac{5}{8}$ " DIA. BOLT - USE SB5/8X24 OR S5TB24 - FOOTING TO BE 30" DEEP.

FOR HOLD DOWNS WITH $\frac{7}{8}$ " DIA. BOLT - USE SB7/8X24 OR S5TB28 - FOOTING TO BE 30" DEEP.

FOR HOLD DOWNS WITH 1" DIA. BOLT - USE SB1X30 - FOOTINGS TO BE 36" DEEP.

slab

shower = 160sqft

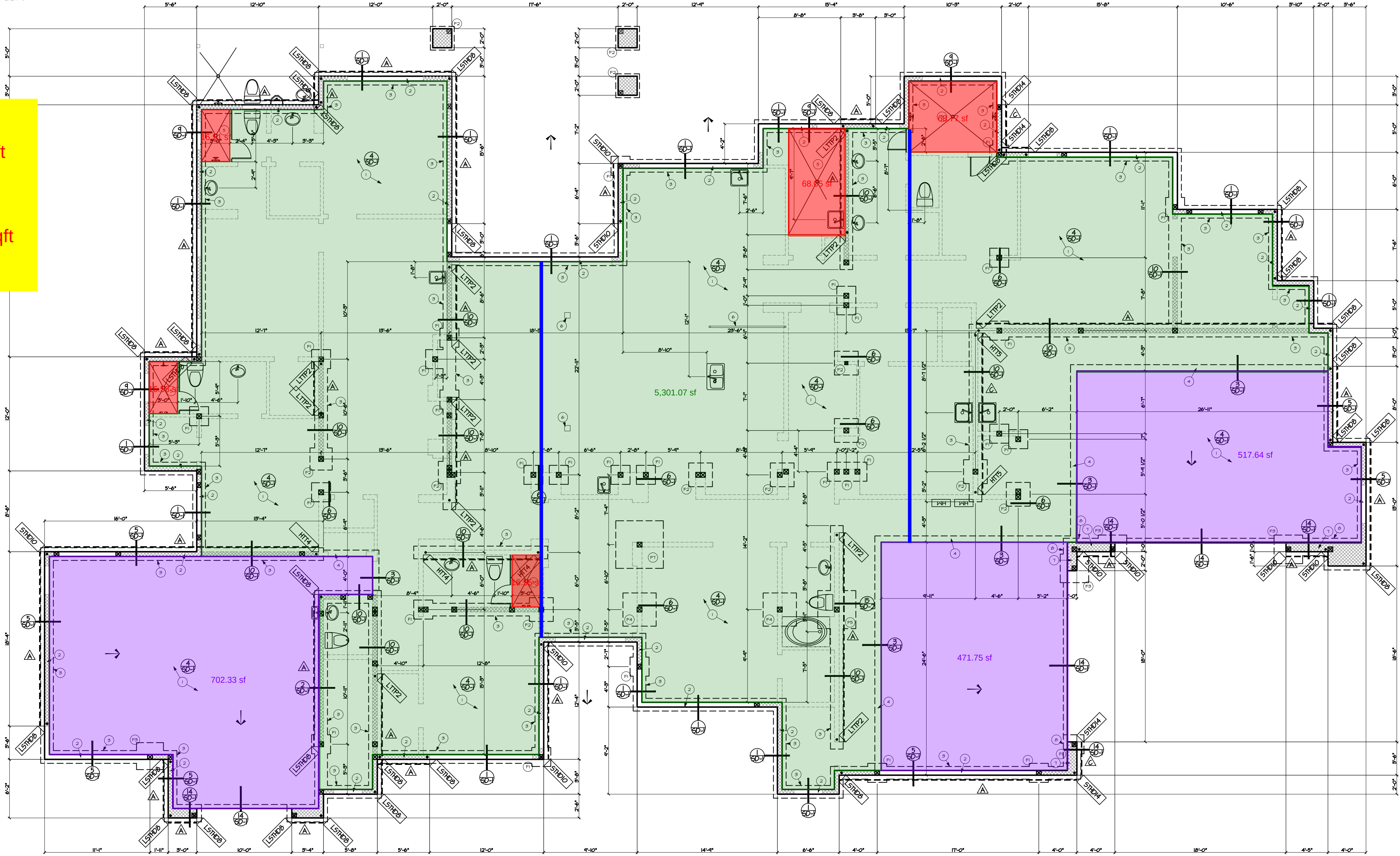
slab = 5300sqft

garage = 1700sqft

KEYNOTES			
* SEE SPECIFICATION FOR MORE INFORMATION *			
* ALL KEYNOTES MAY NOT APPLY TO THIS SHEET *			
No.	DESCRIPTION	No.	DESCRIPTION
①	4" CONCRETE SLAB OVER 4" ABC OVER TERMITES TREATED SOIL.	⑥	RECESSED ELECTRICAL - VERIFY LOCATION WITH OWNER
②	6" STEM WALL WITH #4 HORIZ. AT TOP AND #4 # 48" O.C. VERT. 48 BAR DIA. OVERLAP.	⑦	12" STEM WALL WITH #4 HORIZ. AT TOP AND #4 # 48" O.C. VERT. 48 BAR DIA. OVERLAP.
③	16" WIDE X 12" THICK FOOTING WITH 4-#4 BARS AT BOTTOM OF FOOTING. BOTTOM OF FOOTING 18" BELOW NATIVE SOIL OR ENGINEERED PAD.	⑧	22" WIDE X 10" THICK FOOTING WITH 2-#4 BARS AT BOTTOM OF FOOTING. BOTTOM OF FOOTING 18" BELOW NATIVE SOIL OR ENGINEERED PAD.
④	8" TURNDOWN - SEE DETAILS	⑨	VERIFY WITH CONTRACTOR FOR TRACK DEPTH AT SLAB
⑤	RECESSED SLAB FOR SHOWER DRAIN - SEE DETAILS	⑩	24" #4 BARS AT 32" O.C. - EPOXY WITH SIMPSON SET-XP EPOXY

FOR ALL GENERAL NOTES SEE SHEET G5N	
BOLT LEGEND	
A	1/2" DIA. A.B. X 10" @ 36" O.C., 7" EMB. & 3" SQ WASHER
B	1/2" DIA. A.B. X 10" @ 24" O.C., 7" EMB. & 3" SQ WASHER
C	1/2" DIA. A.B. X 10" @ 18" O.C., 7" EMB. & 3" SQ WASHER
D	1/2" DIA. A.B. X 10" @ 12" O.C., 7" EMB. & 3" SQ WASHER
USE 1/2" DIA. A.B. X 10" @ 48" O.C. UNLESS NOTED OTHERWISE	

FOUNDATION NOTES	
1. FINISH FLOOR SHALL BE A MINIMUM OF 8" ABOVE ADJACENT FINISH GRADE.	7. ALL PROCEDURES, PLACEMENT FORMWORK LAP, ETC. SHALL CONFORM WITH LATEST A.C.I. STANDARDS, MECHANICALLY VIBRATE ALL CONCRETE WHEN PLACED, EXCEPT THE SLABS ON GRADE NEED ONLY TO BE VIBRATED AROUND UNDERFLOOR DUCTS, ETC. MIN. SLUMP 4.5". ALL CONCRETE SLABS ON GRADE SHALL BE BOUNDED BY CONSTRUCTION JOINTS, KEYED OR SANKUT SUCH THAT THE ENCLOSED AREA DOES NOT EXCEED 600 SQUARE FEET. ALL WORK IS TO CONFORM TO ALL LOCAL BUILDING CODES.
2. PROVIDE EXTERIOR WATERPROOF MEMBRANE UP TO 1'-0" BEYOND CONC. STEM AT ALL SUNKEN FLOORS.	8. CONTINUOUS FLOOR SLAB SHALL BE SANKUT EVERY 600 SQUARE FEET FOR EXPANSION. SANKUTS SHALL BE UNDER INTERIOR NON-BEARING WALLS OR IN AREAS NOT AFFECTING TILE FLOORS WHERE POSSIBLE.
3. SEAL ALL VOIDS AROUND PIPING PASSING THRU CONCRETE FLOORS.	9. TOP OF EXTERIOR FOUNDATION SHALL BE 12" ±2% ABOVE THE ELEVATION OF THE STREET GUTTER OR INLET OF AN APPROVED DRAINAGE DEVICE.
4. ALL EXTERIOR CONCRETE FLATWORK TO SLOPE A MINIMUM OF 1/8". FT. AND ALL LANDINGS AT DOORS TO SLOPE A MAXIMUM OF 1/4". FT.	10. ALL BOTTOM PLATES SHALL BE ANCHORED TO THE FOUNDATION WITH 1/2"-ANCHOR BOLTS HAVING 7" MIN. EMBEDMENT @ 4' O.C. THERE SHALL BE A MINIMUM OF TWO BOLTS PER PLATE SECTION WITH ONE BOLT LOCATED NOT MORE THAN 12" OR LESS THAN SEVEN BOLT DIAMETERS FROM EACH END OF THE PLATE SECTION.
5. ALL INTERIOR AND EXTERIOR FOOTINGS SHALL BEAR 18" BELOW ENGINEER CERTIFIED COMPACTED FILL OR UNDISTURBED NATIVE SOIL WITH MINIMUM ALLOWABLE SOIL BEARING PRESSURE AT 1,000 P.S.F.	
6. MIN. 28 DAY COMPREHENSIVE STRENGTH FOR CONCRETE SHALL BE 2500 P.S.I.	



FOUNDATION PLAN
SCALE 3/16" = 1'-0"

PLOTDATE:
CADFILE:

4128 N. 34TH STREET
PHOENIX, AZ 85018
PHONE: (480) 874-7190
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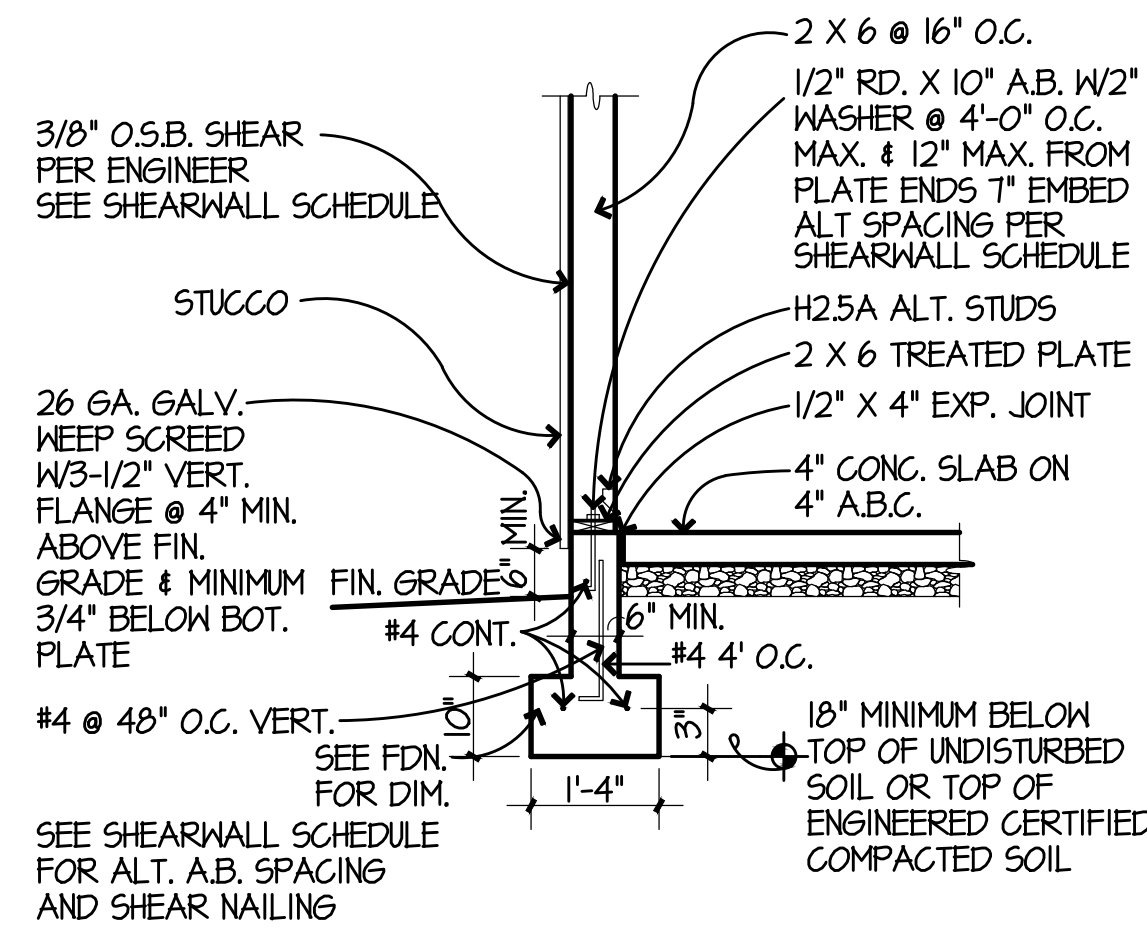
Koch-Haut Res.
9386 S. Krista Dr.
Goodyear, AZ 85338

WILLIAM J. CHRISTENSEN
5/10/2025
REGISTERED PROFESSIONAL ENGINEER
ARIZONA, U.S.A.

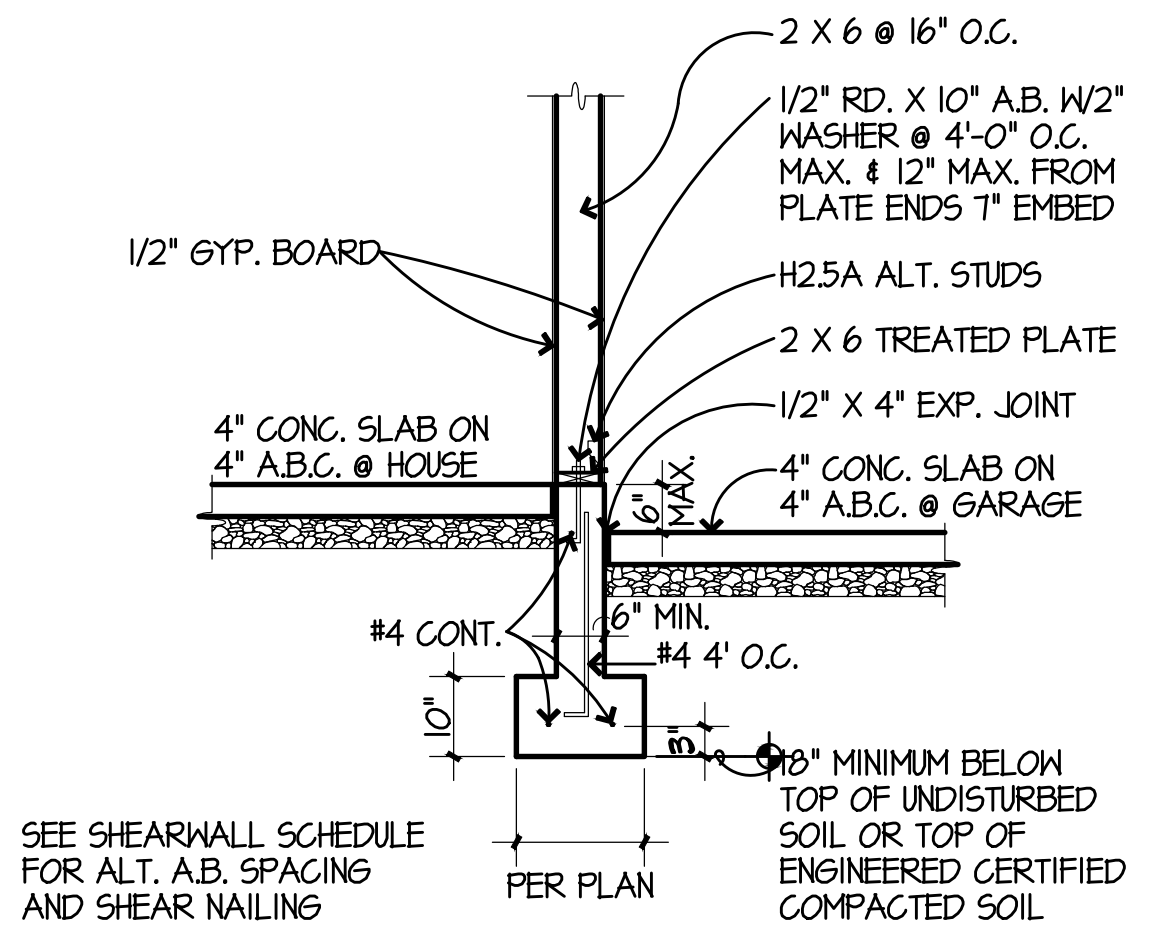
REVISIONS

DATE:	DRAWN BY:	JOB NO.:	REVISION NO.:

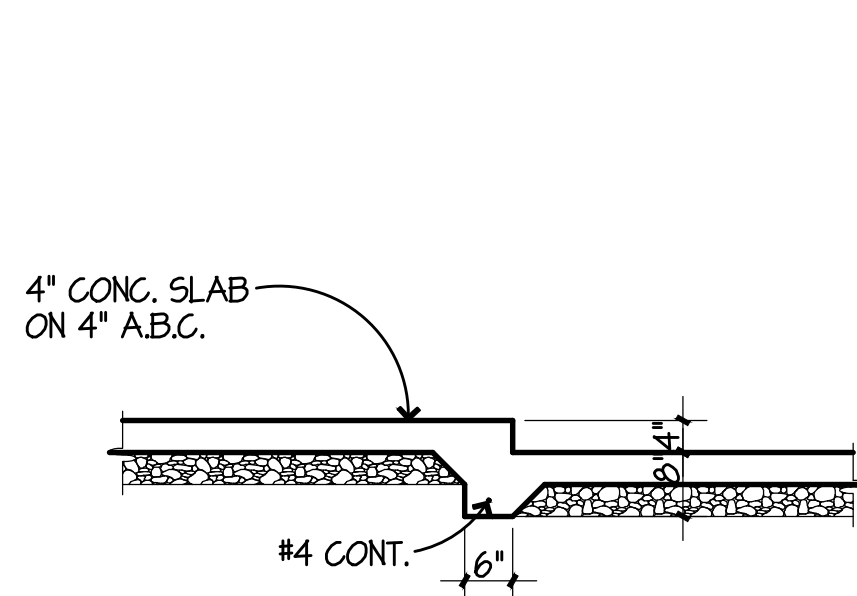
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5-2
OF



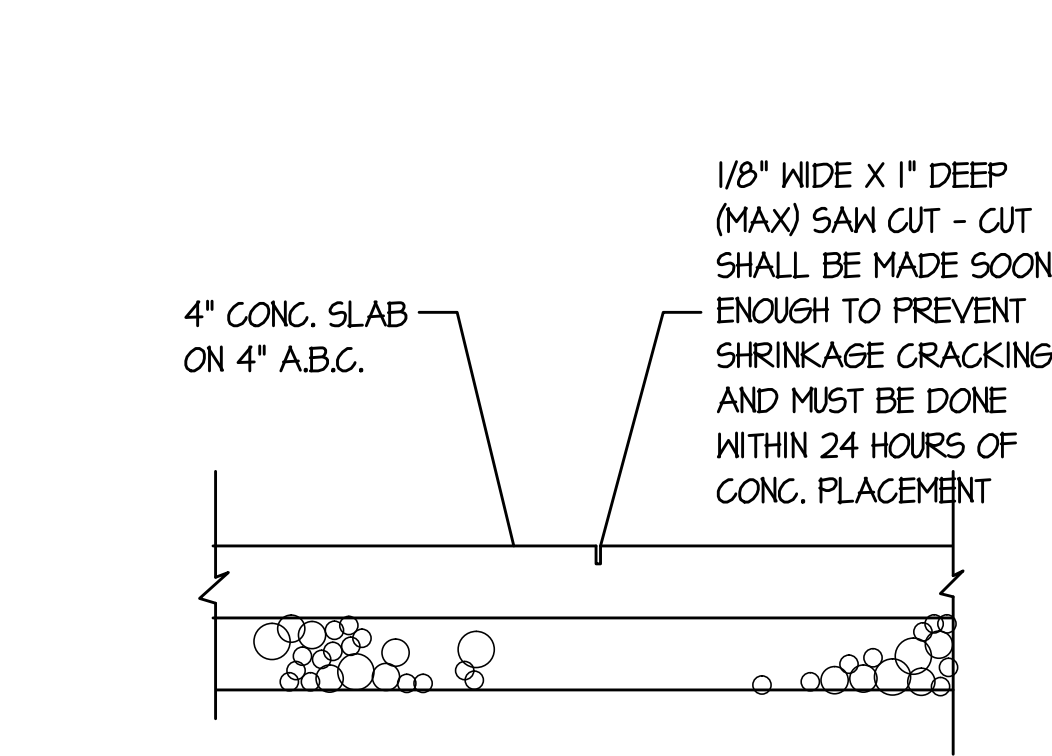
1 TYPICAL STEM



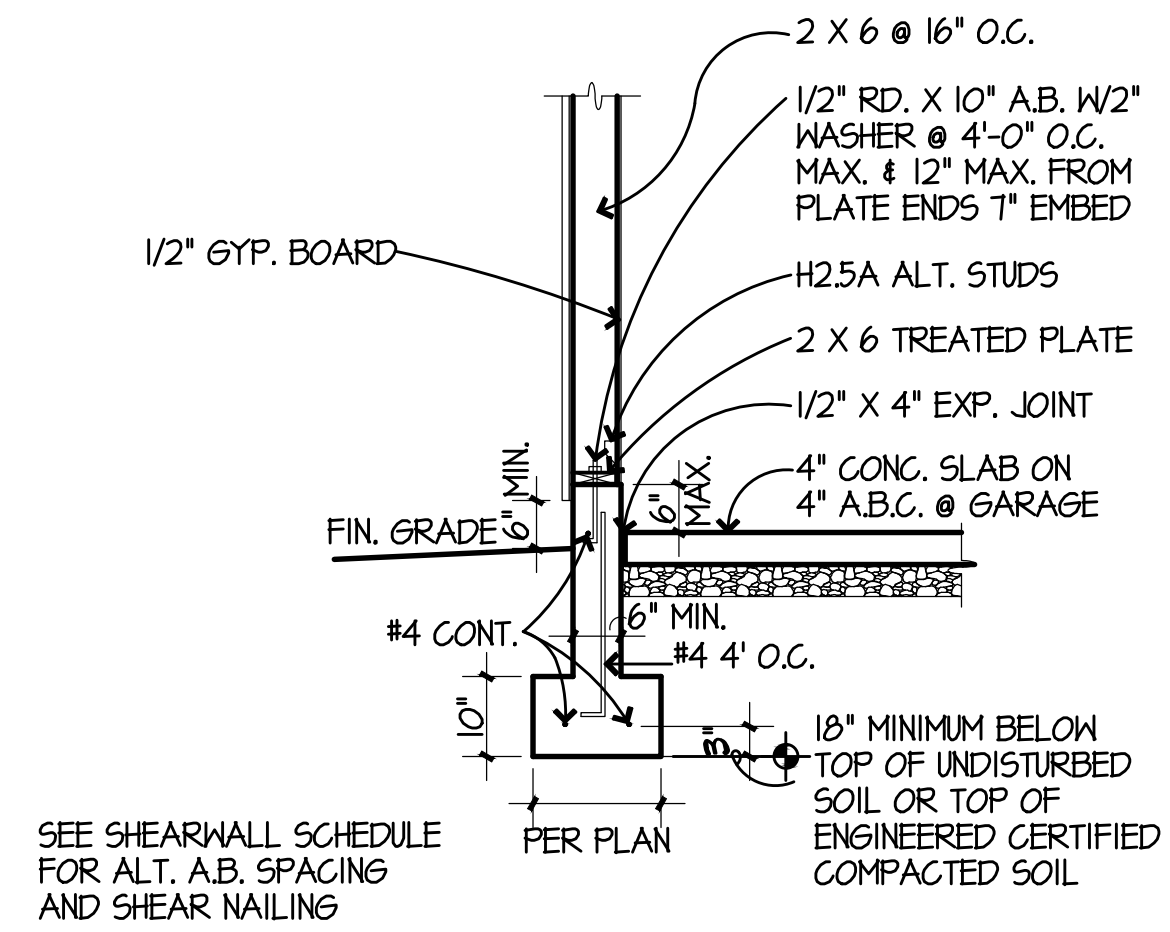
2 FOOTING AT INT. GARAGE WALL



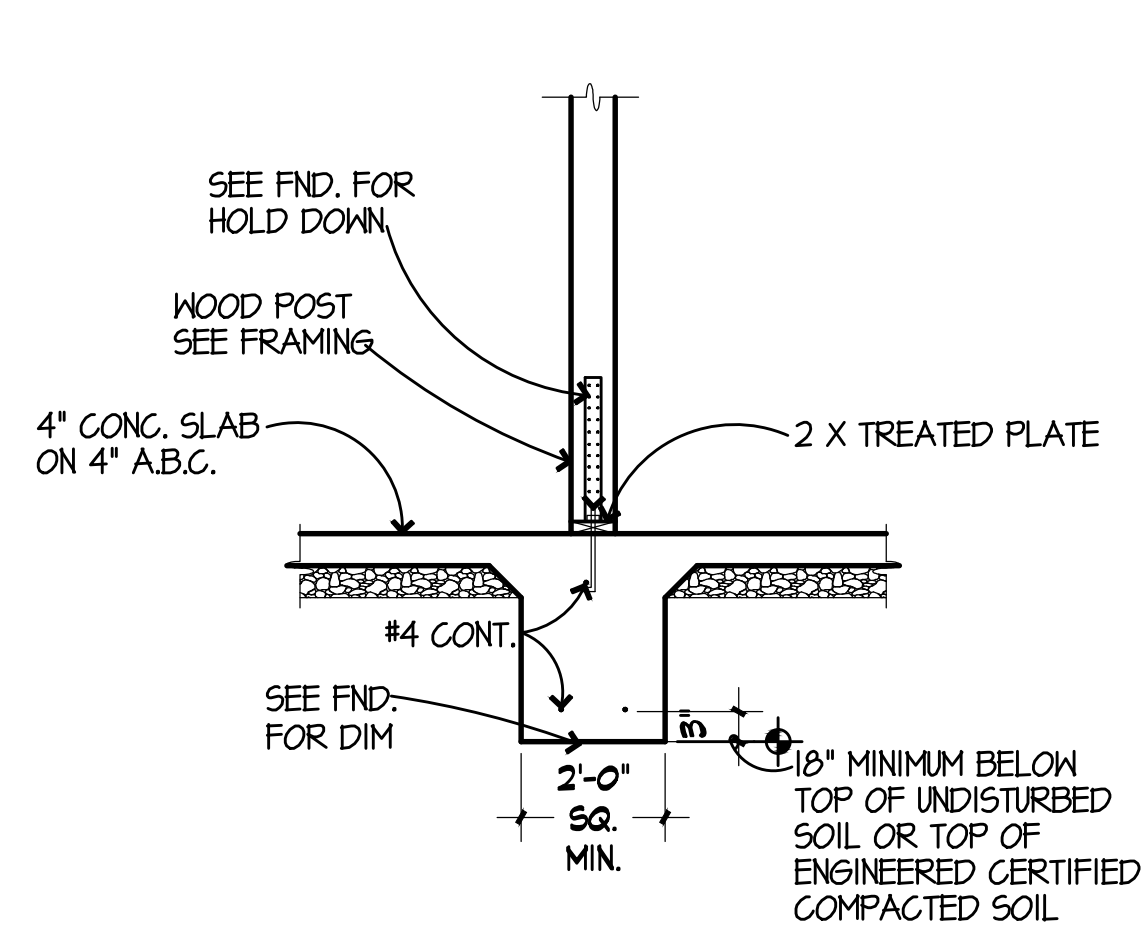
3 FOOTING @ STEPDOWN



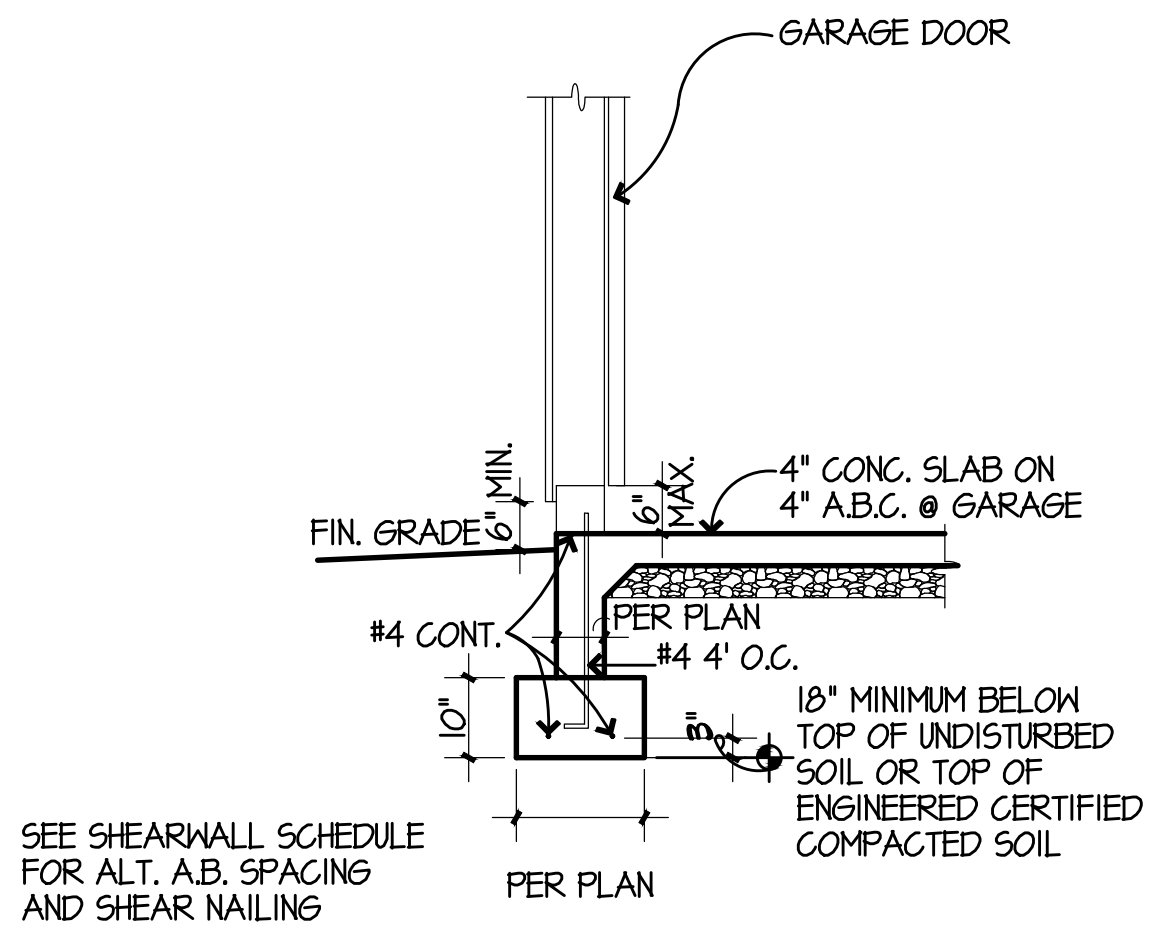
4 CONTROL JOINT



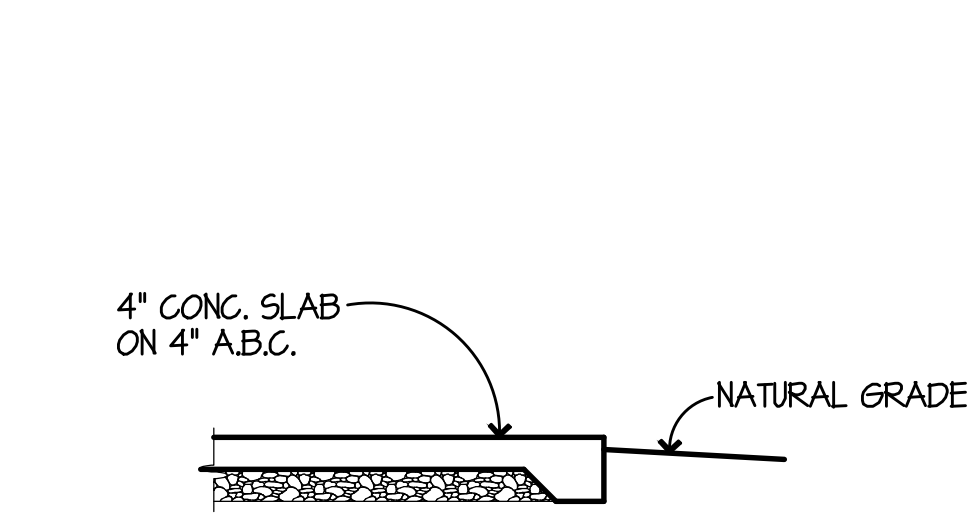
5 FOOTING AT GARAGE WALL



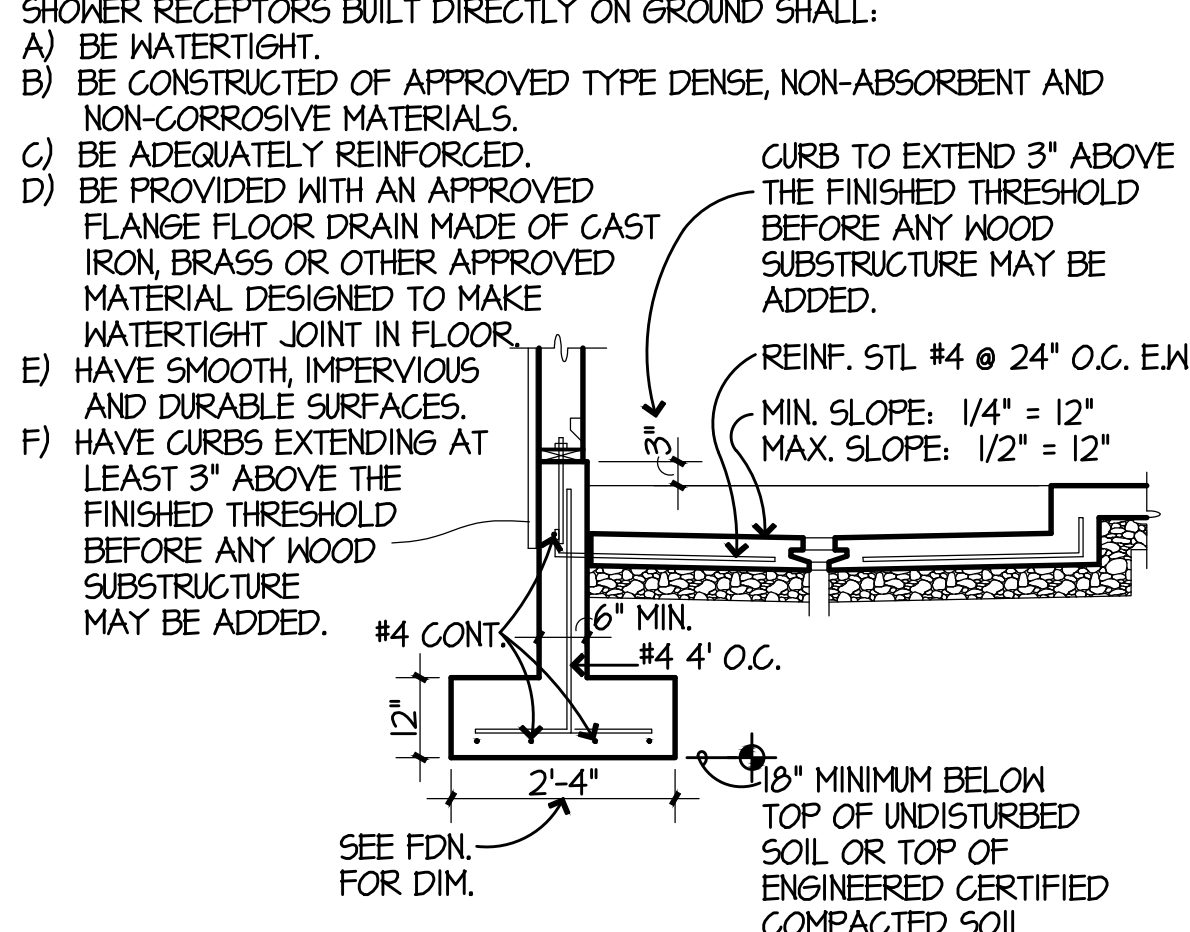
6 INTERIOR PAD FOOTING



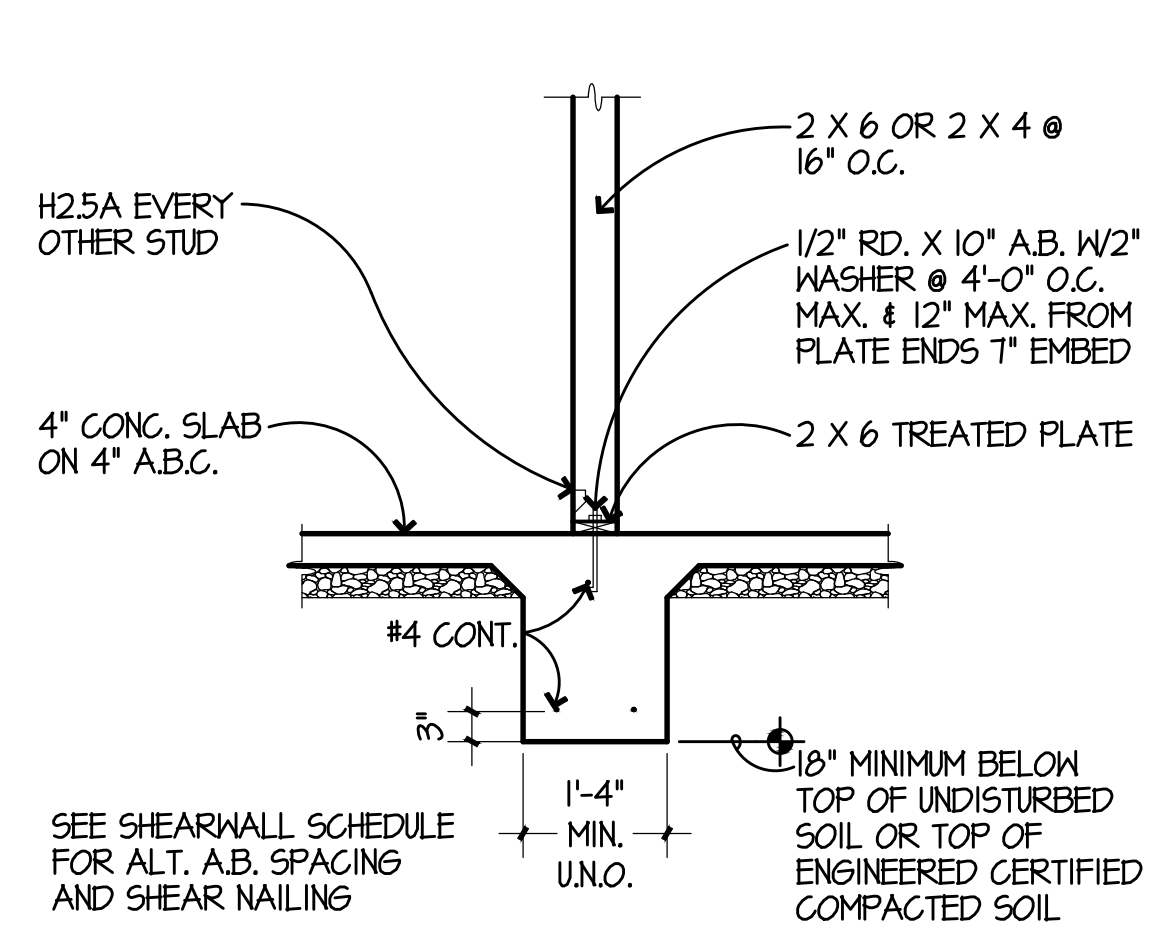
7 FOOTING AT GARAGE DOOR



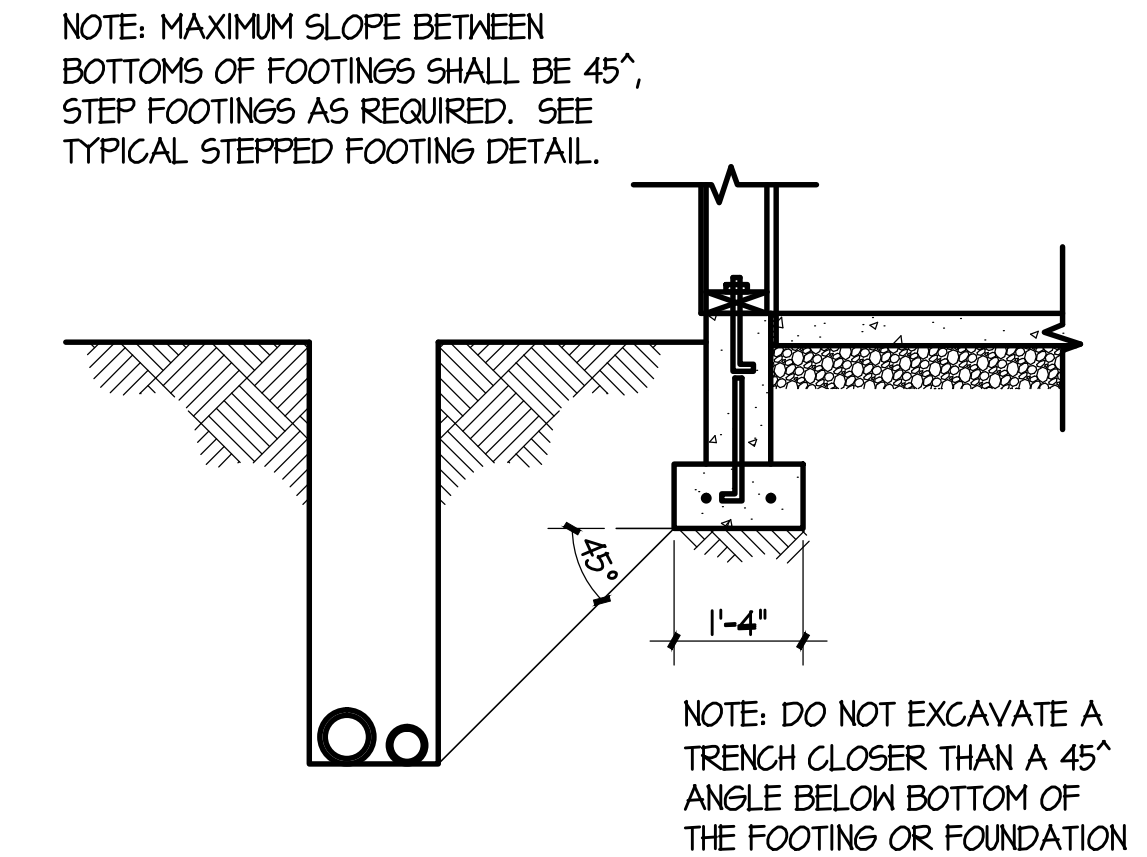
8 SLAB @ PATIO



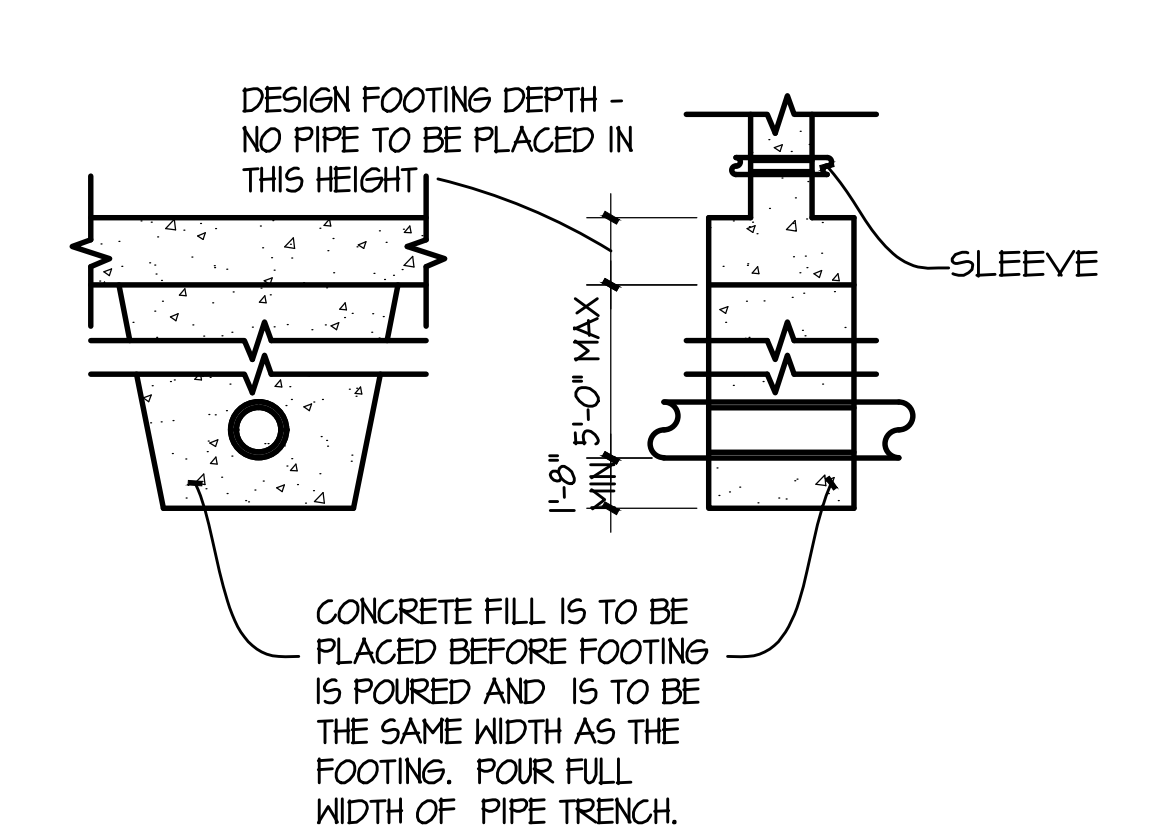
9 FOUNDATION @ SHOWER / LAUN



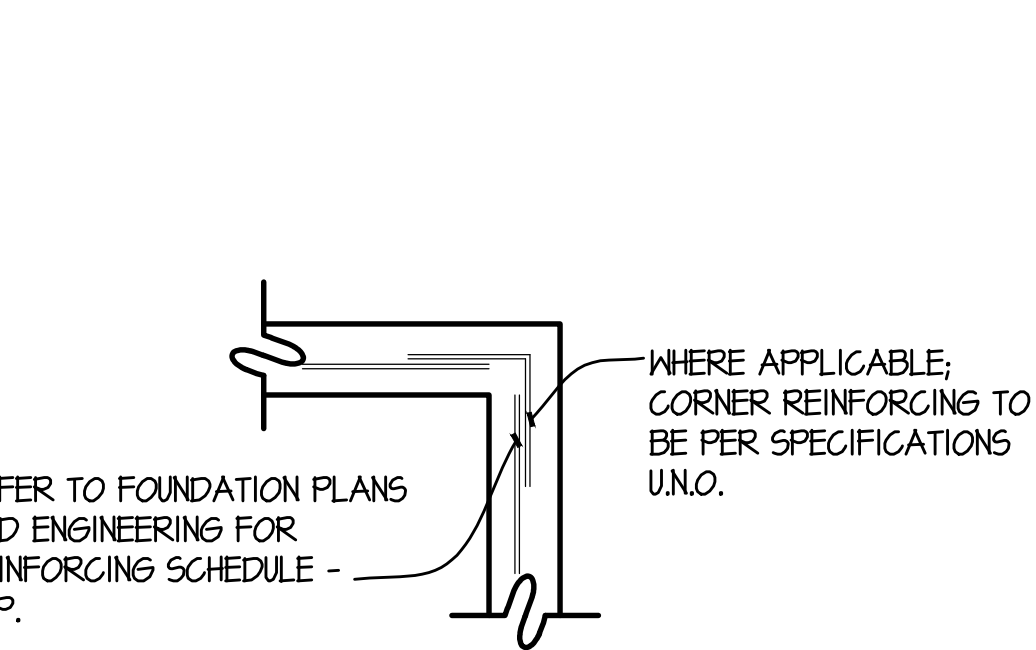
10 INTERIOR BRG. WALL FOOTING



11 TRENCH @ EXTERIOR WALL



12 PIPE THROUGH STEM WALL



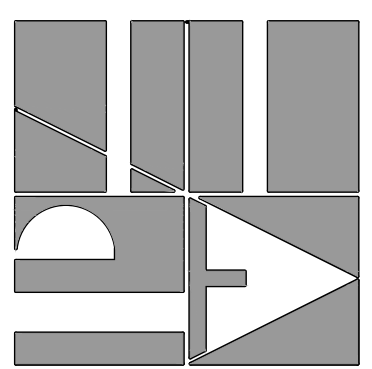
13 CORNER REINFORCING

14

15

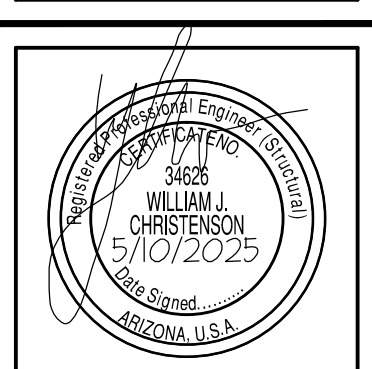
4128 N. 34TH STREET
PHOENIX, AZ 85018
PHONE: (480) 874-7190
FAX: (480) 874-7185

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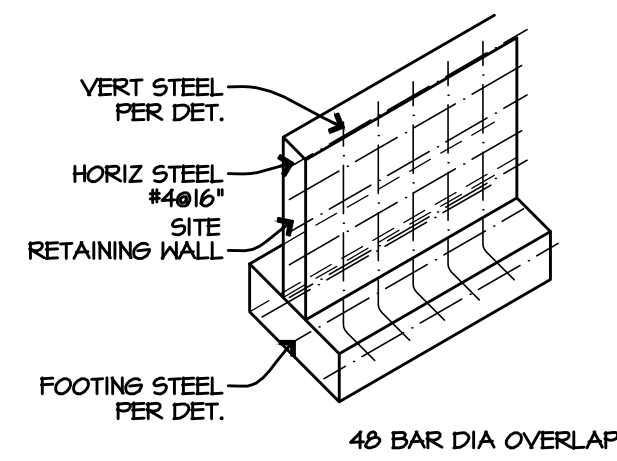
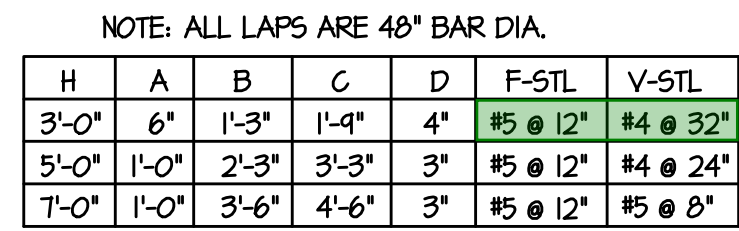
Koch-Haut Res.
9386 S. Krista Dr.
Goodyear, AZ 85338



REVISIONS

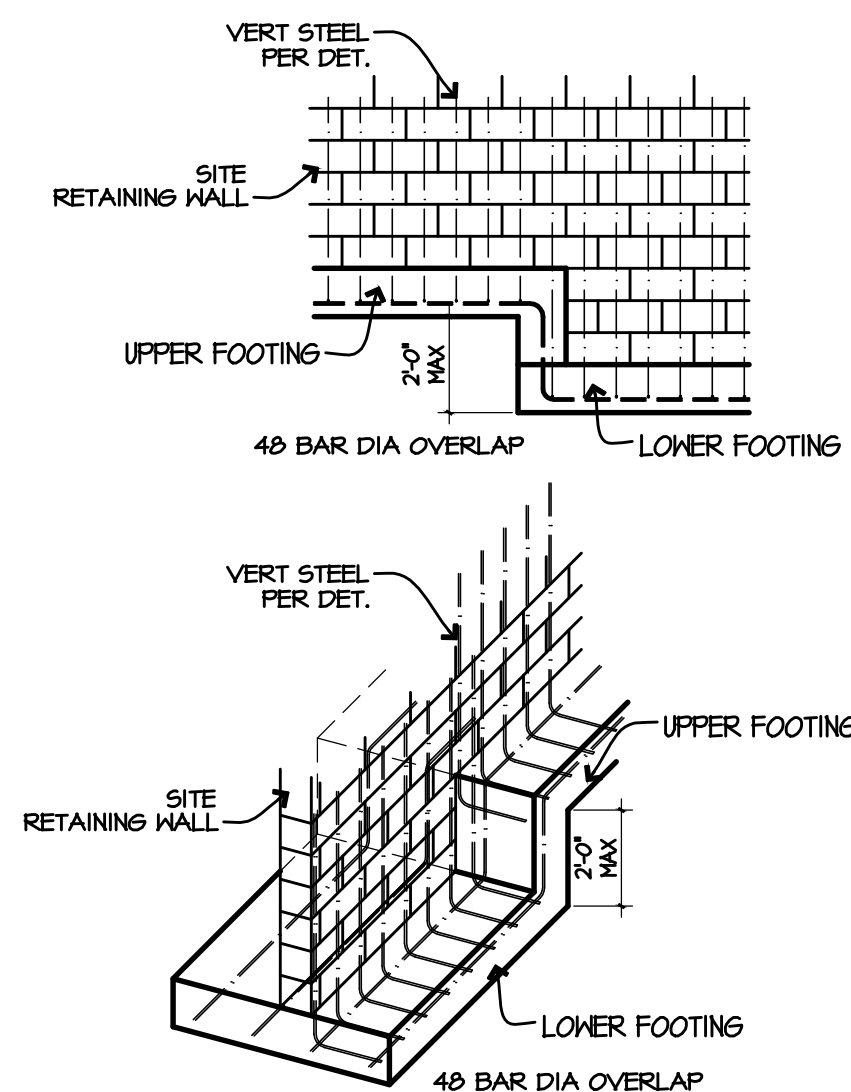
DATE:	DRAWN BY:	JOB NO.:	REVISION NO.:

SHEET NO.
SD-1
OF



BAR OVERLAP	
NOTE: ALL LAPS ARE 48" BAR DIA.	
#3	18"
#4	24"
#5	30"
#6	36"

F_c - 2500 psi
 1500 PSF SOIL BEARING PRESSURE
 220 PSF PASSIVE
 36 ACTIVE
 0.35 SOIL FRICTION
 f_m - 1500 psi
 F_s - 20,000psi
 F_y - 40,000 psi



FOOTING TO BE 3'

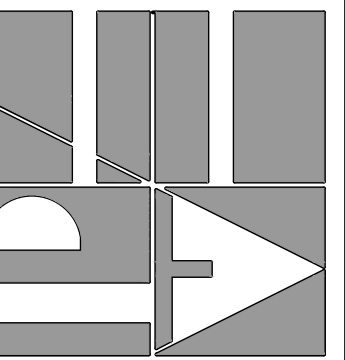
APN: 400-03-151
WAVOMBA EUNICE

RETAINING WALL PLAN

N.T.S.

Patos verdes
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4126 N. 34TH STREET
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CONTRACTOR TO VERIFY ALL MEASUREMENTS IN FIELD.

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REVISIONS
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JOB NO.:
REVISION NO.:

SHEET NO.

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CADFILE: